drones

MDPI

Article

A Hierarchical Cooperative Interception Framework for Multi-UAV Defense Against Large-Scale Swarm Intrusions

Lei Zuo, Ying Wang, Jialu Liu, Yu Lu and Ruiwen Gu *

School of Electronic Engineering, Naval University of Engineering, Wuhan 430033, China; 1309021017@nue.edu.cn (L.Z.); m24385408@nue.edu.cn (Y.W.); m25385407@nue.edu.cn (J.L.); m23183903@nue.edu.cn (Y.L.)

* Correspondence: 1920191143@nue.edu.cn

Highlights

What are the main findings?

- A hierarchical multi-UAV interception framework integrating task allocation, obstacle avoidance, and predictive guidance is proposed, effectively reducing collision risks while improving interception success rates compared to mainstream baseline methods across varying engagement scales.

What are the implications of the main findings?

- The hierarchical interception framework provides a scalable solution for defending against large-scale UAV swarm intrusions in complex urban low-altitude environments, significantly improving task allocation efficiency, flight safety, and interception success rate.

Abstract

To address the challenges of unbalanced task allocation, high inter-UAV collision risks, and lagging interception guidance in multi-UAV cooperative missions within complex urban low-altitude environments, a cooperative interception strategy integrating load-balanced allocation, k-nearest neighbor (k-NN) cooperative obstacle avoidance, and adaptive predictive guidance is proposed. First, a load-balanced Hungarian algorithm is developed at the task allocation layer. The integration of a multi-dimensional distance-angle threat assessment model and a nonlinear load penalty mechanism resolves the issues of resource idling and target overloading inherent in traditional one-to-one allocation, thereby achieving optimal resource configuration for saturated cooperative interception. Second, at the path planning layer, a cooperative obstacle avoidance algorithm based on k-NN nonlinear repulsion is introduced. By exclusively considering the dynamic repulsive fields of local nearest neighbors alongside scale-adaptive parameter regulation, this approach maintains safe formation spacing while reducing the computational complexity from $O(n^2)$ to $O(k)(k \ll n)$, significantly enhancing flight robustness in dense airspaces. Finally, at the terminal guidance layer, an adaptive look-ahead guidance model incorporating motion prediction is constructed to mitigate the overshoot and lag defects associated with classical pure pursuit algorithms during the interception of highly maneuverable targets. The implementation of linear extrapolation and dynamic gain regulation facilitates a paradigm shift from "passive pursuit" to "active interception." Simulation results demonstrate that the proposed algorithm yields substantial improvements in task allocation efficiency, collision risk mitigation, and overall success rates across red-blue UAV swarm confrontation scenarios of varying scales. These findings provide a viable cooperative defense framework against large-scale, highly maneuverable unmanned aerial vehicle (UAV) swarm intrusions.

Check for updates

Academic Editor: Pablo Rodríguez-Gonzálvez

Received: 24 April 2026
Revised: 24 May 2026
Accepted: 27 May 2026
Published: 28 May 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

Keywords: pure pursuit algorithm; task allocation; inter-UAV collision avoidance; UAV interception

1. Introduction

The rapid advancement and widespread deployment of unmanned aerial vehicle (UAV) technology in the civil sector has precipitated increasingly prominent security and regulatory challenges [1]. Notably, unauthorized flights and the ensuing security threats have emerged as critical issues for contemporary social governance and airspace safety, particularly in civilian domains such as nuclear power plants and critical ports, as shown in Figure 1. In complex scenarios involving the simultaneous intrusion of multiple UAVs, traditional countermeasures—such as physical interception nets [2], radio frequency jamming [3], and laser interception [4]—frequently exhibit limitations. These include sluggish response times, restricted coverage areas, susceptibility to environmental interference, and the potential for collateral damage to surrounding electronic equipment [5,6]. Furthermore, conventional methods generally lack sufficient cooperative capabilities and adaptability in dynamic, multi-target environments, rendering them inadequate for achieving efficient, precise, and scalable interception outcomes.

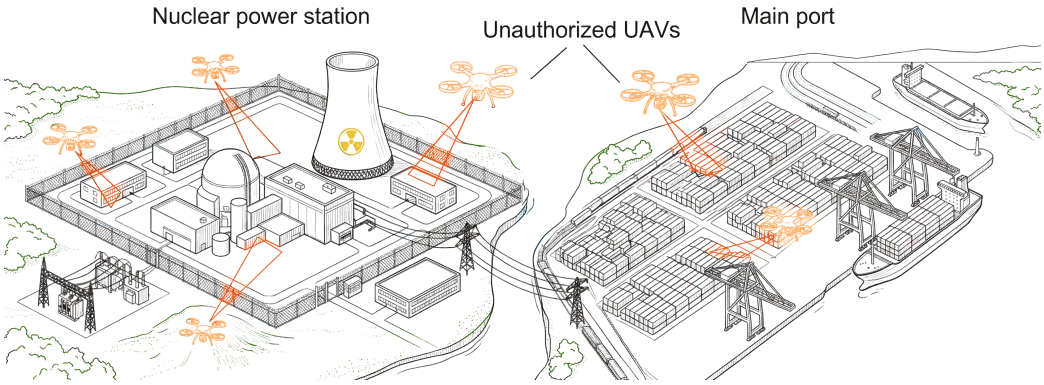


Figure 1. Unauthorized drones flying over nuclear power plants and main ports pose serious threats.

Consequently, recent efforts in both academia and industry have increasingly advocated for a UAV-to-UAV interception paradigm [7,8]. The fundamental advantage of this approach lies in its superior maneuverability, flexibility, and capacity for cooperative engagement. Leveraging autonomous flight and swarm control technologies, interceptor UAVs can execute rapid approaches, continuous tracking, and dynamic responses against intrusive targets. This framework is particularly applicable to complex airspace environments, including urban centers, airports, and critical infrastructure. Compared to static or area-denial defense mechanisms, UAV interception systems offer rapid response, flexible deployment, reusability, and manageable costs. Moreover, through cooperative networking, these systems facilitate regional joint defense and intelligent scheduling, thereby significantly enhancing the operational efficiency and applicability in neutralizing unauthorized UAVs [9,10].

Among the critical technologies underpinning UAV interception systems, interception path planning and guidance algorithms are fundamental to ensuring high strike efficacy. Specifically, the pure pursuit algorithm has emerged as a prominent research focus in interception guidance due to its structural simplicity, low computational overhead, and ease of engineering implementation [11–14]. By continuously adjusting the interceptor's heading such that its velocity vector remains aligned with the instantaneous position of the target, this algorithm demonstrates robust tracking performance in one-on-one engagements

against targets with low maneuverability. However, its limitations become pronounced in complex scenarios involving cooperative intrusions by multiple UAVs. First, multi-UAV cooperative interception missions necessitate the integration of task allocation and path planning algorithms to optimally assign targets, thereby mitigating resource conflicts and task redundancies. Second, interceptors operating in dynamically dense airspaces require real-time inter-vehicle obstacle avoidance capabilities; however, classical pure pursuit algorithms typically lack embedded collision avoidance mechanisms for environmental obstacles and neighboring aircraft. Third, the interception success rate against high-speed, highly maneuverable targets remains limited. Particularly when targets employ evasive maneuvers, the algorithm is susceptible to tracking lag and target loss, culminating in a suboptimal overall interception rate. Consequently, current research predominantly focuses on augmenting the pure pursuit framework through the integration of intelligent methodologies—such as cooperative decision-making [15], predictive guidance [16], differential game theory [17], and reinforcement learning [18]—to enhance the adaptability, robustness, and comprehensive efficacy of multi-target interception systems.

However, as a fundamental algorithmic model, the pure pursuit guidance algorithm can be enhanced within UAV swarm solutions by optimizing obstacle avoidance and cooperative strategies, thereby enabling effective forward jamming or interception. Consequently, the adoption of an improved pursuit algorithm is particularly suitable for small UAVs with limited onboard computing resources in swarms, allowing them to maintain real-time responsiveness even in high-speed maneuvering or dense formation scenarios.

To address the fundamental limitations of the aforementioned classical algorithms in multi-UAV cooperative interception scenarios, algorithmic enhancements are introduced across three dimensions: the task decision layer, the path planning layer, and the terminal guidance layer. The specific design methodology is detailed below:

First, to mitigate inter-UAV resource conflicts and suboptimal cooperative efficiency, an improved load-balanced Hungarian allocation model is designed. Deviating from the conventional single-cost metric based solely on Euclidean distance, this model constructs a comprehensive threat assessment matrix incorporating a distance decay factor and an angular advantage factor. Concurrently, the introduction of a nonlinear load penalty term circumvents the rigid one-to-one constraint of the standard Hungarian algorithm, thereby realizing a saturated and well-balanced allocation of interception resources at the mathematical programming level.

Second, to address inter-vehicle collision risks during formation flight in dense airspaces, a cooperative obstacle avoidance mechanism based on the K-nearest neighbor principle is proposed. In contrast to the traditional artificial potential field method, which suffers from high global computational overhead and local minima entrapment, this mechanism restricts the repulsive force influence strictly to the local K nearest neighbors. By incorporating a nonlinear repulsive function that dynamically adjusts to the formation scale, this approach ensures microscopic safe flight separation while facilitating smooth macroscopic interception trajectories with significantly reduced computational demands.

Third, to overcome the tracking lag and target loss induced by the highly maneuverable evasive actions of targets, an adaptive pure pursuit guidance algorithm integrating target motion prediction is developed. By embedding a linear extrapolation predictor and a relative-distance-based dynamic look-ahead gain within the terminal guidance loop, the interceptor's heading reference is shifted from the target's instantaneous position to a predicted interception point. This modification effectively mitigates the inherent deficiencies of classical pure pursuit algorithms, notably the inefficiency of tail-chasing trajectories and severe terminal overshoot.

Moreover, parameter configuration is a critical aspect of multi-UAV system design. Recent advances in dynamical systems optimization, such as the genetic algorithm-based stability analysis framework [19], have demonstrated the effectiveness of evolutionary computation in automatically tuning system parameters to meet stability and performance constraints. The integration of such parameter optimization techniques into the present hierarchical framework is further discussed in Section 6.

It should be noted that the simulation framework in this study adopts the following assumptions: (1) Each UAV is modeled as a point-mass kinematic model with a fixed-step update ($\Delta t = 0.1$ s), disregarding physical factors such as mass, inertia, and wind resistance. Consequently, the heading of each UAV can change instantaneously without turn-rate or acceleration constraints at the guidance level, representing an upper-bound performance benchmark. (2) The defense coordinator possesses perfect and instantaneous global awareness of all attacking drones' positions and velocities. (3) A centralized communication architecture is assumed: the defense coordinator broadcasts the position of every red UAV and every friendly blue UAV to all defending drones via a perfect, zero-latency, infinite-bandwidth link at each time step, and each defending drone receives this information without loss or delay. The coordinator also possesses velocity information for all red UAVs; however, this velocity data is utilized exclusively at the task allocation layer (Section 2) for trajectory prediction, while the terminal guidance layer relies solely on position information received at each time step. (4) Each defending drone has perfect self-localization capability (equivalent to ideal GPS/IMU integration with no sensor noise). (5) Attacking drones have zero situational awareness of the defenders. They execute pre-programmed cubic Bézier trajectories at constant speed without any reactive, evasive, or cooperative behavior. (6) No onboard sensors are modeled—perfect state information of both friendly and hostile aircraft is available to all decision-making modules. These assumptions are widely adopted in the existing literature on multi-UAV cooperative interception and task allocation [20–23], and they allow the study to focus on the algorithmic properties of the proposed hierarchical framework without conflating the results with low-level dynamics or sensor noise.

The organization of this paper is as follows, as shown in Figure 2. Section 1 serves as the introduction, elaborating on the research background and current challenges. Section 2 details the cooperative task allocation mechanism based on a load-balanced Hungarian algorithm, primarily encompassing multi-dimensional threat assessment modeling and nonlinear penalty function design. Section 3 focuses on an inter-UAV cooperative obstacle avoidance strategy within the formation, presenting a comparative analysis of the performance differences among a traditional artificial potential field (APF) method, an APF with an escape mechanism (improved APF), and a proposed K-nearest neighbor (K-NN) nonlinear repulsion model (cooperative k-NN). Section 4 presents an improved terminal guidance algorithm; utilizing the classical pure pursuit algorithm as a baseline, it derives a cooperative predictive interception law that incorporates adaptive look-ahead and target prediction capabilities. Section 5 provides a comprehensive performance summary of the hierarchical interception framework, consolidating the multi-metric evaluation results from task allocation, obstacle avoidance, and terminal guidance layers. Finally, the discussion (Section 6) and conclusions (Section 7) are outlined.

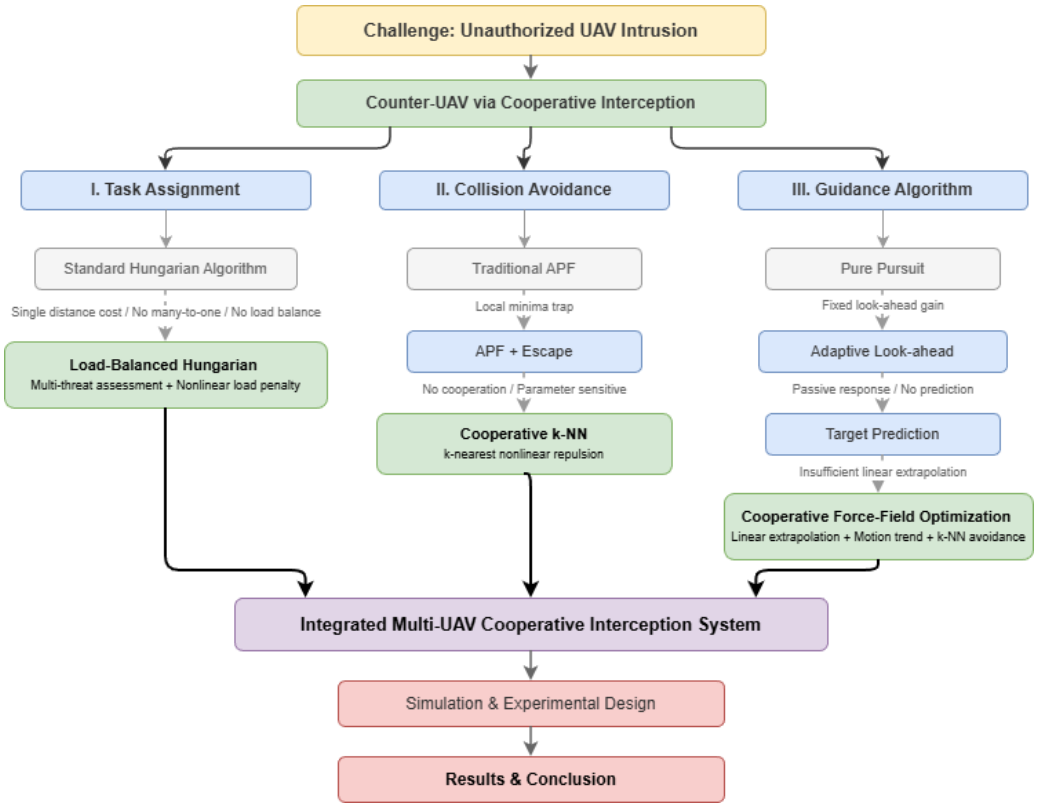


Figure 2. The framework of this paper.

2. Task Allocation Algorithm

2.1. Hungarian Algorithm

In cooperative unmanned aerial vehicle (UAV) interception missions, the efficient allocation of n blue UAVs to m red targets constitutes a quintessential combinatorial optimization problem. The core challenge lies in reconciling the fairness of task allocation with overall interception efficiency, while simultaneously satisfying stringent real-time constraints. The Hungarian algorithm [24], also known as the Kuhn–Munkres algorithm, is a classical method for solving optimal bipartite matching problems. Proposed in 1955, this algorithm iteratively optimizes the matching scheme by identifying augmenting paths, ultimately achieving a global optimum within an $O(n^3)$ time complexity.

Let the set of blue UAVs be denoted as $B = \{b_1, b_2, \ldots, b_n\}$, and the set of red targets as $R = \{r_1, r_2, \ldots, r_m\}$. Define the cost matrix as $C \in \mathbb{R}^{n \times m}$, where c_{ij} represents the cost of allocating blue UAV b_i to red target r_j. The standard Hungarian algorithm formulates this problem as the following integer linear programming model:

$$\min \sum_{i=1}^{n} \sum_{j=1}^{m} c_{ij} \cdot x_{ij} \tag{1}$$

constraints:

$$\begin{aligned} &\sum_{j=1}^{m} x_{ij} = 1, \quad \forall i \in \{1, 2, \ldots, n\} \\ &\sum_{i=1}^{n} x_{ij} \leq 1, \quad \forall j \in \{1, 2, \ldots, m\} \\ &x_{ij} \in \{0, 1\}, \quad \forall i, j \end{aligned} \tag{2}$$

where $x_{ij} = 1$ denotes the assignment of blue UAV b_i to red target r_j. Existing research typically employs Euclidean distance as the cost metric:

$$\begin{aligned} c_{ij} &= \| \mathbf{p}_i^b - \mathbf{p}_j^r \| \\ &= \sqrt{\left(x_i^b - x_j^r\right)^2 + \left(y_i^b - y_j^r\right)^2 + \left(z_i^b - z_j^r\right)^2} \end{aligned} \tag{3}$$

where $\mathbf{p}_i^b = (x_i^b, y_i^b, z_i^b)$ and $\mathbf{p}_j^r = (x_j^r, y_j^r, z_j^r)$ denote the position vectors of the i-th blue interceptor and the j-th red intruder in the three-dimensional inertial frame, respectively, and $\| \cdot \|$ represents the Euclidean (L2) norm.

However, the application of the standard Hungarian algorithm to UAV interception scenarios reveals significant limitations that necessitate targeted improvements. Specifically, the absence of a many-to-one allocation mechanism precludes the execution of saturated allocation. Moreover, the algorithm relies exclusively on a singular distance-based cost evaluation, neglecting critical factors such as engagement angles and threat levels. Specifically, when $n > m$, the one-to-one constraint in Equation (2) leaves $n - m$ blue UAVs unassigned, resulting in idle defense resources despite an overall numerical advantage. Meanwhile, the reliance on a single distance-based cost metric fails to account for engagement angles and threat levels, further limiting tactical optimality.

To mitigate these inherent defects, the academic community has proposed several modified schemes. For instance, coalition formation algorithms relax the rigid one-to-one constraint, permitting the assignment of multiple blue UAVs to a single red target [25,26]; while this enhances resource utilization, it fails to overcome the limitations associated with the single-distance cost metric. Alternatively, receding horizon optimization algorithms operate by periodically recalculating allocations, which facilitates the correction of early erroneous decisions but still leaves the load-balancing issue unresolved [27]. Furthermore, heuristic approaches, such as dung beetle optimizer and so on, are capable of managing multi-objective optimization [28]; however, their substantial computational overhead renders them unsuitable for real-time operational scenarios.

2.2. Load-Balanced Hungarian Algorithm

To address the aforementioned limitations, a Load-Balanced Hungarian Algorithm is proposed. The core principle entails achieving an equilibrium between fairness and efficiency in task allocation through the incorporation of a multi-dimensional threat assessment and a nonlinear load penalty mechanism.

1. Integrated Threat Assessment Model:
   (a) Distance Threat Component: A rational decay model is adopted, with sensitivity dynamically adjusted according to scale. It is defined as follows:

$$\begin{aligned} Threat_{dist}(i,j) &= \frac{1}{1 + d_{ij} \cdot \alpha_{scale}} \\ \alpha_{scale} &= \begin{cases} 0.115, & n \le 20, \\ 0.210, & 20 < n \le 35, \\ 0.360, & n > 35. \end{cases} \end{aligned} \tag{4}$$

   where $d_{ij} = \|\mathbf{p}_i^b - \mathbf{p}_j^r\|$ denotes the Euclidean distance between the i-th blue UAV and the j-th red target (as defined in Equation (3)), and $\alpha_{scale} > 0$ represents the scale-adaptive distance sensitivity factor, which directly governs the decay rate of the distance threat score, $Threat_{dist}$. The value of α_{scale} is assigned piecewise based on the total number of blue UAVs, n: for small-scale formations ($n \le 20$), it is set to 0.115 to prevent premature locking onto a single target and preserve global response flexibility; for medium-scale formations ($20 < n \le 35$), it is

set to 0.210; and for large-scale formations ($n > 35$), it is established at 0.360 to prioritize proximal threats, reduce interception distances, and mitigate task conflicts. The aforementioned thresholds, determined via parameter scanning in preliminary experiments, consistently balance load distribution and interception timeliness within the range of $n \in [20, 60]$.

(b) Angular Threat Component: Considering the orientation advantage of the UAV, it is defined as follows:

$$Threat_{angle}(i,j) = \frac{1}{1 + \theta_{ij} \cdot 0.5} \tag{5}$$

where $\theta_{ij} (0^\circ \leq \theta_{ij} < 180^\circ)$ represents the angle between the global reference direction (positive X-axis, pointing toward the intruder side) and the relative position vector from blue UAV i to red target j. The angular factor of 0.5 is an empirically selected compromise designed to provide moderate differentiation for angular variance while preventing excessive penalties on off-axis targets, thereby maintaining the core allocation principle of "distance priority."

(c) Integrated Threat Score: A weighted fusion strategy is defined as follows:

$$\begin{gathered} Threat_{score}(i,j) = w_{dist} \cdot Threat_{dist}(i,j) + w_{angle} \cdot Threat_{angle}(i,j) \\ (w_{dist}, w_{angle}) = \begin{cases} (0.85, 0.15) & n \leq 20, \\ (0.75, 0.25) & 20 < n \leq 35, \\ (0.90, 0.10) & n > 35. \end{cases} \end{gathered} \tag{6}$$

where w_{dist} and w_{angle} are the fusion weights for distance threat and angular threat, respectively, satisfying the constraint $w_{dist} + w_{angle} = 1$. The weight ratio is adaptively adjusted based on the blue formation scale n: small-scale scenarios ($n \leq 20$) utilize $(0.85, 0.15)$, where distance remains the dominant factor and angle serves as a secondary metric to maintain geometric rationality of interception paths in sparse airspace; medium-scale scenarios ($20 < n \leq 35$) utilize $(0.75, 0.25)$, moderately increasing the angular weight to address entry-angle competition under multi-UAV coordination; and large-scale scenarios ($n > 35$) utilize $(0.90, 0.10)$, further reinforcing the distance-priority principle to compel the system to rapidly suppress proximal high-risk targets in high-density environments, thereby reducing computational and decision-making delays associated with angular considerations. These piecewise weights are determined through preliminary experimental optimization to maintain a dynamic equilibrium between response efficiency and allocation rationality across different scales.

Figure 3 illustrates how the load-balancing algorithm is applied to the actual UAV allocation process. By comprehensively considering both distance threat and angular threat, the integrated threat score between each blue UAV and each red target is determined, and optimization allocation is performed based on this score. Here, θ_{ij} represents the relative angle between blue UAV i and red target j, and d_{ij} denotes the distance between them. For example, in the figure, $\theta_{11} > \theta_{12}$ and $d_{11} > d_{12}$ indicate that red target 2 has a smaller engagement angle and shorter distance relative to blue UAV 1, yielding $Threat_{\text{score}}(1,1) < Threat_{\text{score}}(1,2)$. Consequently, blue UAV 1 is prioritized for assignment to the higher-threat red target 2.

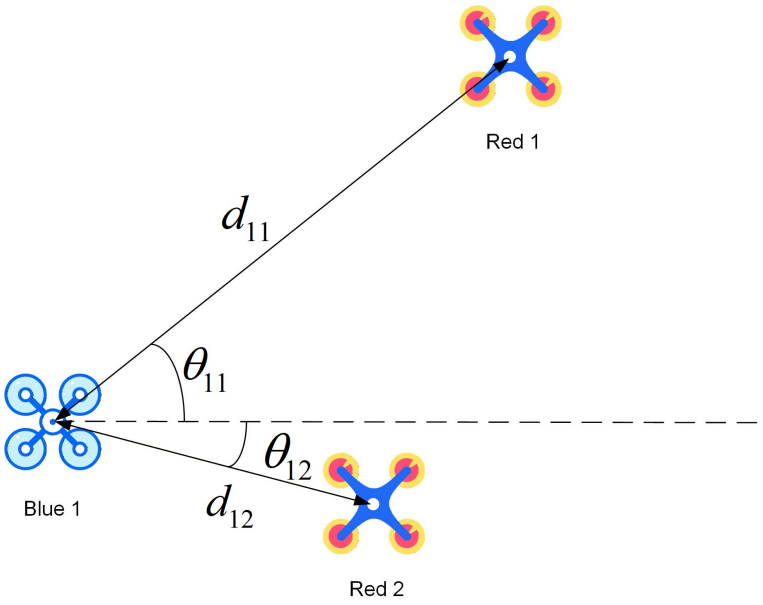


Figure 3. Schematic diagram of the load-balancing algorithm.

2. Nonlinear Load Penalty Mechanism:

   (a) Let $Load_j \in \mathbb{N}_0$ denote the number of blue UAVs already assigned to red target r_j during the current allocation phase (i.e., the current load). The load penalty function is defined as follows:

$$Penalty_{load}(j) = \lambda_{scale} \cdot (Load_j)^{1.2} \tag{7}$$

   where $\lambda_{scale} \in \mathbb{R}^+$ denotes the scale-adaptive penalty coefficient; and the exponent 1.2 serves as a nonlinear adjustment factor. This exponent is set slightly higher than a linear term (1.0) but significantly lower than a quadratic term (2.0) to achieve a specific effect: providing moderate penalty growth during low-load phases to facilitate many-to-one cooperative interception, while accelerating penalty increments during high-load phases to suppress excessive resource stacking on a single target (The exponent 1.2 is determined empirically. The many-to-one allocation is designed for redundancy when interceptors outnumber targets, without assuming a probabilistic single-engagement success model). The value of λ_{scale} is assigned piecewise based on the total number of blue UAVs, n: for small-scale formations ($n \leq 20$), it is set to 0.006, as a lightweight penalty is sufficient to maintain balanced allocation when interceptors are relatively abundant; for medium-scale formations ($20 < n \leq 35$), it is set to 0.038 to moderately increase the penalty in response to intensifying target competition; and for large-scale formations ($n > 35$), it is established at 0.070 to further reinforce the load repulsion effect, preventing local target overloading and defensive vacuums in high-density scenarios. These coefficients were determined via parameter scanning in preliminary experiments to stably achieve an optimal compromise between load balancing and interception efficiency within the range of $n \in [20, 60]$.

   (b) To align with the physical intuition of interception scenarios, the assignment problem is reformulated as a utility maximization problem. The interception utility function, u_{ij}, is defined to comprehensively measure the target's threat value against the current load cost:

$$u_{ij} = Threat_{score}(i, j) - Penalty_{load}(Load_j) \tag{8}$$

The higher the utility value, the greater the net gain in assigning blue UAV b_i to red target r_j. For the blue UAV b_i currently awaiting assignment, the red target r_j that maximizes the net gain is selected as follows:

$$j^* = \arg \max_{j \in \{1,2,\ldots,m\}} u_{ij} \tag{9}$$

For the i-th blue UAV currently awaiting assignment, the optimal target index is denoted as j^*.

Figure 4 demonstrates this nonlinear load penalty mechanism, which successfully suppresses excessive resource accumulation by dynamically penalizing the number of already assigned blue UAVs on the same target, thereby achieving a more balanced task allocation. For instance, in the figure, the red target Red 1 is assigned to the blue UAV Blue 1. At this point, when blue UAVs Blue 2 and Blue 3 evaluate the allocation utility of Red 1, in addition to considering distance and angular threats, they are also affected by the penalty from the current load. As Blue 1 is assigned, the load on Red 1 increases, leading to a decrease in its utility value. Consequently, Blue 2 and Blue 3 are more inclined to select other targets (such as Red 2), achieving rational resource dispersion and load balancing.

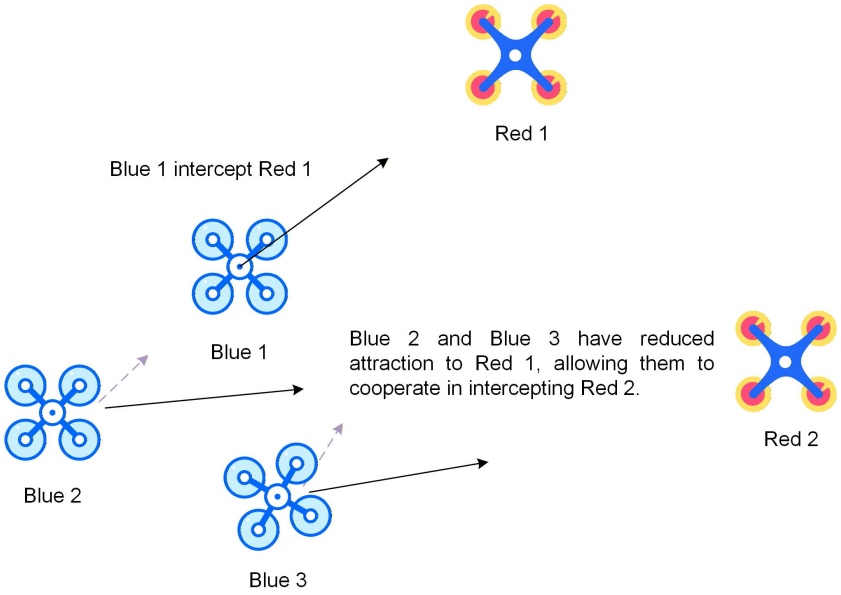


Figure 4. Load penalty mechanism.

To independently evaluate the effectiveness of the proposed load-balanced allocation strategy, the lower-level obstacle avoidance and guidance modules were fixed to baseline configurations in this section. The simulation environment used in all three sections shares the following configuration: blue UAVs are initialized at ground level and assemble into a circular formation; red intruder trajectories are defined by cubic Bézier curves in a wedge formation. Blue and red speeds are 25 m/s and 30 m/s, respectively. The task allocation algorithm was treated as the sole independent variable in comparative experiments conducted across four engagement scales (15v20, 30v40, 40v50, and 50v60). To eliminate the influence of stochastic disturbances, each configuration was executed for 50 independent trials. An iterative Hungarian algorithm, which repeatedly applies the standard one-to-one Hungarian matching to the remaining unassigned blue UAVs until all are allocated, served as the comparative baseline. The evaluation metrics encompassed four dimensions: load standard deviation, maximum-to-minimum load ratio, total flight distance, and average computational latency. The experimental results are summarized in Table 1 and Figure 5.

Table 1. Task Assignment Performance Comparison.

Scale	Algorithm	Load Std	Max-Min Ratio	Distance (m)	Time (ms) [1]
15v20	Hungarian [2]	1.291	6.000	9242.14	0.274
	LB-Hungarian	**0.488**	**2.000**	**9430.13**	**0.410**
30v40	Hungarian	1.826	11.000	21,284.03	0.752
	LB-Hungarian	**0.479**	**2.000**	**22,058.53**	**1.232**
40v50	Hungarian	1.581	11.000	29,535.56	1.240
	LB-Hungarian	**0.439**	**2.000**	**30,309.44**	**2.059**
50v60	Hungarian	1.414	11.000	39,038.56	2.028
	LB-Hungarian	**0.404**	**2.000**	**39,808.42**	**3.080**

[1] Algorithm computation time. [2] An iterative greedy variant: the standard one-to-one Hungarian matching is repeatedly applied to the remaining unassigned blue UAVs until all are allocated. Bold: highlight the proposed method.

Regarding load balancing, the LB-Hungarian algorithm demonstrates a significant advantage; its load standard deviation (Load Std) remains within the range 0.4–0.5 across all scales, outperforming the 1.4–1.8 range of the standard Hungarian algorithm, as shown in Figure 5a. This indicates that the former achieves a more uniform distribution of blue UAVs among red targets, effectively reducing the phenomenon where certain targets are overloaded while others remain idle. The maximum-to-minimum load ratio (Max-Min Ratio) further corroborates this advantage: LB-Hungarian consistently maintains a ratio at an ideal level of 2.0, signifying that the most heavily loaded target is assigned exactly twice as many UAVs as the least loaded one, as shown in Figure 5b. In contrast, this metric reaches as high as 6-11 for the standard algorithm, reflecting extreme load imbalance where a multitude of UAVs may cluster on a few targets while others receive virtually no coverage.

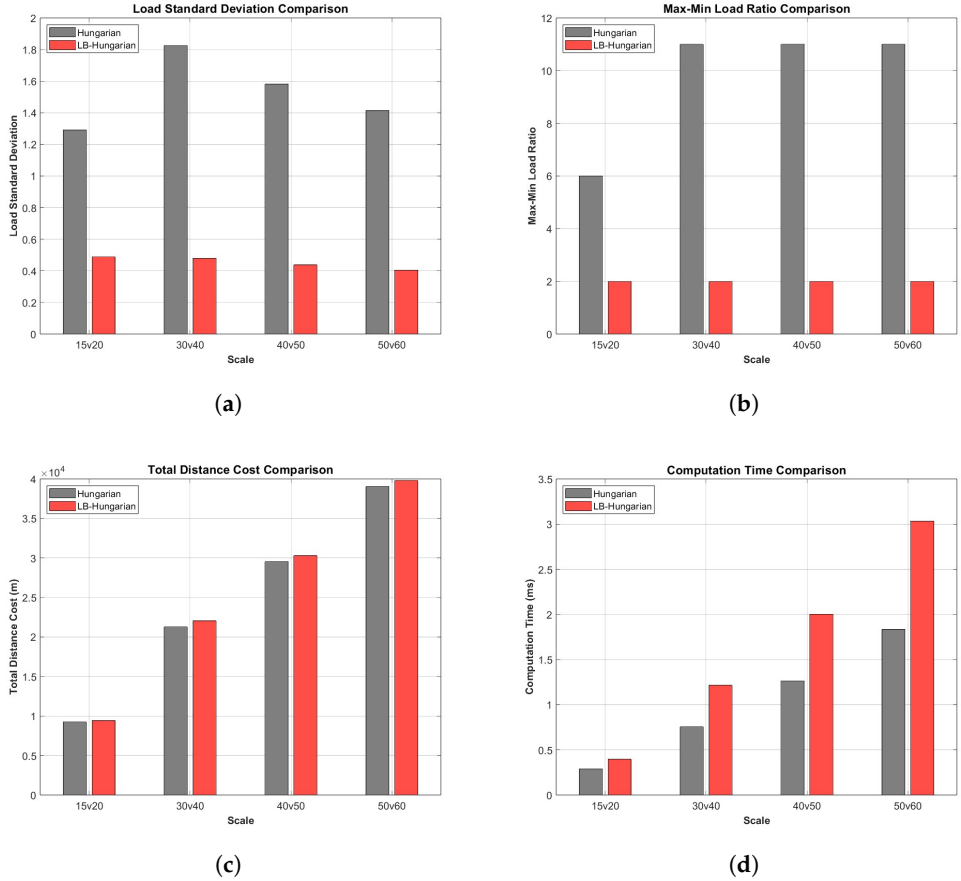


Figure 5. (**a**) Load standard deviation comparison. (**b**) Max–Min load ratio comparison. (**c**) Total distance cost comparison. (**d**) Computation time comparison.

In terms of total flight distance, the standard algorithm maintains a slight advantage, with values approximately 2–3% lower than those of LB-Hungarian, as shown in Figure 5c. This is expected, as the standard algorithm performs greedy allocation based purely on distance minimization. However, this marginal gain comes at the cost of severe load disparity; LB-Hungarian sacrifices minimal distance efficiency to secure a substantial improvement in load balancing—a trade-off that is tactically justified in combat scenarios. Regarding computation time, LB-Hungarian is slightly higher than the standard algorithm (approximately 1.5–1.8 times), as shown in Figure 5d, due to the increased complexity of calculating comprehensive threat scores and load penalties. Nevertheless, even at the 50v60 scale, the per-allocation computation time remains approximately 3 ms on an AMD Ryzen 7 255 CPU (tested using MATLAB 2024b), fully satisfying real-time operational requirements. Taken together, these data clearly illustrate that the LB-Hungarian algorithm significantly improves load-balancing performance while maintaining acceptable computational overhead, achieving an effective unification of many-to-one cooperative interception and equitable resource allocation.

While more complex task allocation algorithms exist, the primary objective of this section is not to advance the theoretical frontiers of combinatorial optimization, but rather to provide a computationally efficient and load-aware upstream module for the downstream obstacle avoidance and guidance layers. The comparison with the standard Hungarian algorithm sufficiently demonstrates the necessity of load balancing in saturated interception scenarios.

3. Inter-UAV Obstacle Avoidance Algorithm

In cooperative UAV interception missions, ensuring safe obstacle avoidance within the formation while maintaining interception efficiency constitutes a critical tactical challenge. The primary difficulty lies in effectively mitigating the risk of inter-vehicle collisions during dense formation flight without introducing excessive interception delays or increased miss distances. To isolate the comparative analysis of obstacle avoidance algorithms and eliminate interference from high-order dynamic coupling, this study utilizes a point-mass kinematic model and a pure pursuit guidance framework as a unified trajectory control baseline. Within this configuration, all potential field force vectors are normalized into desired heading unit vectors, while velocity tracking is independently executed by the low-level controller.

3.1. Artificial Potential Field

The Artificial Potential Field (APF) method, proposed by Khatib in 1986 [29], is a seminal approach to robotic path planning. By simulating physical potential fields, the algorithm treats the target point as an attractive pole and obstacles as repulsive poles, thereby guiding the robot's movement under the influence of the resultant potential force.

The traditional artificial potential field method treats the UAV as a particle moving within a potential field. For the i-th blue UAV, let its current position be denoted by $\mathbf{p}_i^b$ and the position of the assigned interception target by $\mathbf{p}_g$. The UAV is subjected to the combined effects of an attractive force $\mathbf{F}_{att}$ directed toward the target and a repulsive force $\mathbf{F}_{rep}$ originating from neighboring aircraft. The attractive force is defined as:

$$\mathbf{F}_{att} = \xi \cdot (\mathbf{p}_g - \mathbf{p}_i^b) \tag{10}$$

where ξ is the attraction factor, controlling the strength of the pull toward the target. And the repulsive force is defined as:

$$\mathbf{F}_{rep} = \eta \cdot \sum_{i=1}^{n} \left(\frac{1}{d_i} - \frac{1}{d_0} \right)^2 \cdot \frac{1}{d_i^2} \cdot \nabla d_i \tag{11}$$

where η denotes the repulsive factor, d_i represents the distance between the host vehicle and the i-th neighboring aircraft, d_0 is the repulsive influence radius, and ∇d_i is the distance gradient (directed from the host vehicle toward the neighbor). The summation is performed over all friendly aircraft within the range defined by d_0. (In this section, the attraction factor $\xi = 1.0$, and the repulsion factor $\eta = 20.0$, ensuring that close-range obstacle avoidance takes priority; $d_0 = 12m$.)

Then, the total potential field force acting on the UAV is the vector sum of the attractive and repulsive forces:

$$\mathbf{F}_t = \mathbf{F}_{att} + \mathbf{F}_{rep} \tag{12}$$

Therefore, the desired heading direction for the UAV is obtained by normalizing the total force vector:

$$\mathbf{d_t} = \frac{\mathbf{F}_t}{\| \mathbf{F}_t \|} \tag{13}$$

However, the standard APF exhibits significant limitations when applied to UAV formation obstacle avoidance scenarios, necessitating targeted improvements. The main drawbacks of traditional APF include: local minima traps, lack of cooperative obstacle avoidance mechanisms, and parameters that cannot adapt to different scales. To address these issues, the academic community has proposed various improvement schemes, primarily including local minima escape mechanisms and obstacle avoidance strategies based on cooperative optimization.

3.2. Improved Artificial Potential Field Method

To address the local minima problem inherent in the traditional APF, the APF + Escape (improved APF) algorithm incorporates an escape strategy to actively disengage from potential field traps upon detecting a stagnant state. Distinct from conventional methods, this study employs a wall-following strategy [30] rather than the virtual repulsive source approach [31].

Specifically, the escape triggering mechanism jointly evaluates the following two types of indicators:

$$\begin{aligned} &\frac{1}{10} \sum_{i=t-9}^{t} \|\mathbf{v}(i)\| < v_{threshold} = 0.5 \text{ m/s} \\ &\sigma\left(\frac{\mathbf{F}(t)}{\| \mathbf{F}(t) \|} \right) > \sigma_{thresh} = 0.3 \end{aligned} \tag{14}$$

where $\mathbf{v}(i)$ is the UAV velocity at step i; $v_{threshold} = 0.5$ m/s is the speed stagnation bound, and this threshold corresponds to 2% of the nominal blue UAV cruising speed (25 m/s), reliably indicating local minimum entrapment; $\mathbf{F}(t)$ denotes resultant force vector acting on the UAV at the current time step t, synthesized from the attractive and repulsive potential fields; $\sigma(\cdot)$ is the standard deviation taken over the preceding 10 normalized force directions; and $\sigma_{thresh} = 0.3$ is the oscillation detection threshold. The two thresholds are empirically determined values: $v_{threshold} = 0.5$ m/s is used to determine whether the UAV has lost effective forward progress, while $\sigma_{thresh} = 0.3$ is utilized to detect severe oscillations in the direction of the resultant force. Preliminary experiments demonstrate that these values exhibit stable triggering characteristics across various formation scales.

Once stagnation is detected, the algorithm activates a wall-following strategy:

$$\mathbf{d}_{escape} = \mathbf{d}_{\perp} \tag{15}$$

and:

$$\mathbf{d}_{\perp} = \pm \frac{\mathbf{d}(t) \times \mathbf{F}(t)}{\| \mathbf{d}(t) \times \mathbf{F}(t) \|} \tag{16}$$

where $\mathbf{d}_{\perp}$ denotes the unit escape direction orthogonal to both the current heading $\mathbf{d}(t)$ and the resultant force $\mathbf{F}(t)$; a sign variable $s \in \{+, -\}$ is introduced to select the direction that maximizes alignment with the target unit direction $\mathbf{d}_g$, i.e.,

$$\arg \max_{s \in \{\pm\}} s \cdot (\mathbf{d}(t) \times \mathbf{F}(t)) \cdot \mathbf{d}_g \tag{17}$$

thereby ensuring the escape trajectory preserves progress toward the assigned target, as shown in Figure 6. In the figure, the total force on Blue 1 at step i is $\mathbf{F}(\mathbf{t})$, and its flight direction is $\mathbf{d}(\mathbf{t})$. Both $\mathbf{F}(\mathbf{t})$ and $\mathbf{d}(\mathbf{t})$ lie within plane Plan1. Plane Plan2 is orthogonal to plane Plan1. The two vectors $+\mathbf{d}_{\perp}$ and $-\mathbf{d}_{\perp}$ shown are both orthogonal to $\mathbf{d}(\mathbf{t})$ and $\mathbf{F}(\mathbf{t})$. However, the angle $\varphi 1$ formed between $+\mathbf{d}_{\perp}$ and the currently assigned target direction $\mathbf{d}_g$ is smaller than $\varphi 2$ (the angle between $-\mathbf{d}_{\perp}$ and $\mathbf{d}_g$). Therefore, $\mathbf{d}_{\perp} = +\mathbf{d}_{\perp}$ is chosen to maintain flight stability.

The wall-following strategy effectively guides the UAV to navigate around the obstacle by following its contour, which is particularly beneficial in scenarios where the UAV is trapped between multiple obstacles or in narrow passages. By continuously monitoring the escape state and adjusting the heading accordingly, the algorithm can successfully extricate the UAV from local minima and resume effective interception trajectories.

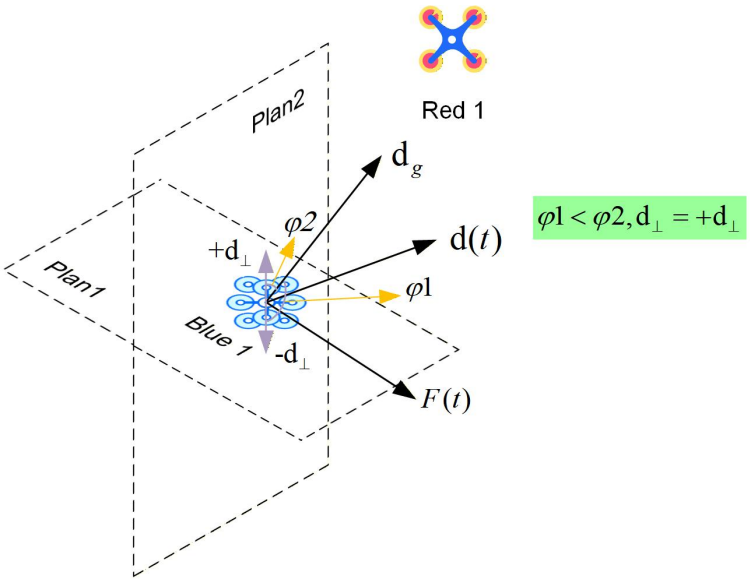


Figure 6. Wall-following strategy escape scheme.

This approach effectively mitigates the local minima problem, facilitating disengagement from potential field traps while preserving the inherent simplicity of the APF algorithm through the mere addition of an escape detection module.

However, it incurs increased computational overhead due to the requirement for continuous monitoring of velocity and escape states, and the time complexity remains $O(n^2)$. Furthermore, it fails to address the underlying cooperative obstacle avoidance challenge, as it focuses exclusively on individual optimization and exhibits high parameter sensitivity. Critically, in scenarios with a large number of targets, the stochastic nature of the escape maneuvers may introduce secondary oscillations.

3.3. K-Nearest Neighbor Cooperative Obstacle Avoidance

To address the limitations of the traditional APF and APF + Escape methods, this study proposes the Cooperative *k*-Nearest Neighbor (Cooperative *k*-NN) obstacle avoidance algorithm, which introduces two core improvements:

- k-Nearest Neighbor Repulsion: The algorithm considers repulsive forces only from the k nearest neighbors, thereby avoiding the $O(n^2)$ complexity associated with global calculations.
- Dynamic Parameter Adaptation: The value of k, the repulsion influence radius, and the intensity parameters are dynamically adjusted based on the total number of UAVs. This ensures that the algorithm maintains effective obstacle avoidance performance across different formation scales without requiring manual parameter tuning.

For the i-th blue UAV, let its current position be denoted by $\mathbf{p}_i^b$, and the set of positions of the remaining friendly UAVs be defined as:

$$\mathcal{P}_{-i} = \{\mathbf{p}_j^b \mid j = 1, 2, \ldots, n. (j \neq i)\} \tag{18}$$

First, calculate the Euclidean distances between the host UAV and each friendly UAV as below:

$$d_{ij} = \| \mathbf{p}_j^b - \mathbf{p}_i^b \|, \quad j \in \mathcal{P}_{-i} \tag{19}$$

where the distances are sorted in ascending order, and the friendly aircraft corresponding to the first k minimum distances are selected as the set of nearest neighbors $\mathcal{N}_k^{(i)}$. Denote these distances in ascending order, with corresponding position vectors $\mathbf{p}_1^b, \mathbf{p}_2^b, \ldots, \mathbf{p}_k^b$.

Based on this local neighbor information, construct a nonlinear repulsive force direction vector:

$$\mathbf{d}_{avoid} = -\sum_{m=1}^{k} \left(1 - \frac{d_m}{d_r}\right)^{\alpha} \cdot \lambda \cdot \frac{\mathbf{p}_m^b - \mathbf{p}_i^b}{d_m} \tag{20}$$

where $d_r > 0$ represents the threshold for the repulsive force influence range, $\alpha = 2.5$ is the nonlinear decay exponent, and $\lambda > 0$ is the repulsive intensity coefficient. Notably, the summation $\sum_{m=1}^{k}$ only iterates over the k nearest neighbors, rather than all $n-1$ friendly aircraft. In the traditional APF method, each UAV must compute repulsive contributions from all other $n-1$ UAVs, yielding a per-UAV computational cost of $O(n)$ and a total cost of $O(n^2)$ for the entire formation. In contrast, the proposed Cooperative *k*-NN method reduces the per-UAV cost to $O(k)$, resulting in a total complexity of $O(k \cdot n)$. Since $k \ll n$ (in this study, $k \in \{3, 4, 5\}$ while $n \in [15, 60]$, as shown in Equation (22)), this design achieves a significant reduction from $O(n^2)$ to approximately $O(n)$, enabling real-time computation in dense formations. Next, the raw repulsive vector is normalized to obtain the unit obstacle avoidance direction:

$$\mathbf{d}'_{avoid} = \frac{\mathbf{d}_{avoid}}{\|\mathbf{d}_{avoid}\|} \tag{21}$$

this mechanism restricts the calculation of repulsive forces to the local k nearest neighbors, ensuring safe inter-vehicle separation while significantly reducing computational overhead in dense formations. Furthermore, the repulsion range d_r, the number of neighbors k and the repulsive intensity coefficient λ can be adaptively adjusted according to the formation scale, allowing the system to meet obstacle avoidance requirements across varying airspace densities. It defines the parameter adaptation scheme as follows:

$$(k, d_r, \lambda) = \begin{cases} (4, 6.0, 0.40), & n \leq 20, \\ (3, 5.5, 0.35), & 20 < n \leq 35, \\ (5, 6.5, 0.42), & n > 35. \end{cases} \tag{22}$$

In practical applications, Cooperative k-NN employs a simplified two-weight synthesis strategy, which linearly combines the base interception direction with the *k*-nearest neighbor obstacle avoidance direction:

$$\mathbf{d}_f = (1 - w_{avoid}) \cdot \mathbf{d}_g + w_{avoid} \cdot \mathbf{d}_{avoid} \tag{23}$$

Then, the final desired heading direction $\mathbf{d}_f^{'}$ is obtained by normalizing the synthesized vector $\mathbf{d}_f$. $w_{avoid} \in [0, 1]$ is the obstacle avoidance weight coefficient. The value of w_{avoid} is adaptively adjusted based on the formation scale: for small-scale formations ($n \leq 20$), it is set to 0.40 to provide moderate obstacle avoidance influence while maintaining a strong focus on the target; for medium-scale formations ($20 < n \leq 35$), it is reduced to 0.35 to balance the increased risk of inter-vehicle collisions with the need for efficient interception; and for large-scale formations ($n > 35$), it is increased to 0.42 to further enhance obstacle avoidance capabilities in dense airspace, where the likelihood of close encounters is significantly higher. These values are determined through parameter scanning in preliminary experiments, ensuring that the algorithm maintains effective obstacle avoidance performance across different scales while minimizing negative impacts on interception efficiency.

In Figure 7, the left section shows the force analysis diagram for Blue 1. In both the traditional APF and improved APF algorithms, the forces acting on it originate from other UAVs. Specifically, the area circled in red, taking the position of Blue 6 as an example, may become a deadlock escape direction for Blue 1, leading to further deadlock. This effect stems from the uncertainty introduced by its escape behavior. The right section of the figure employs the k-nearest neighbor obstacle avoidance algorithm for computation. Using Blue 1 as an example for force analysis again, the computational time complexity is significantly reduced.

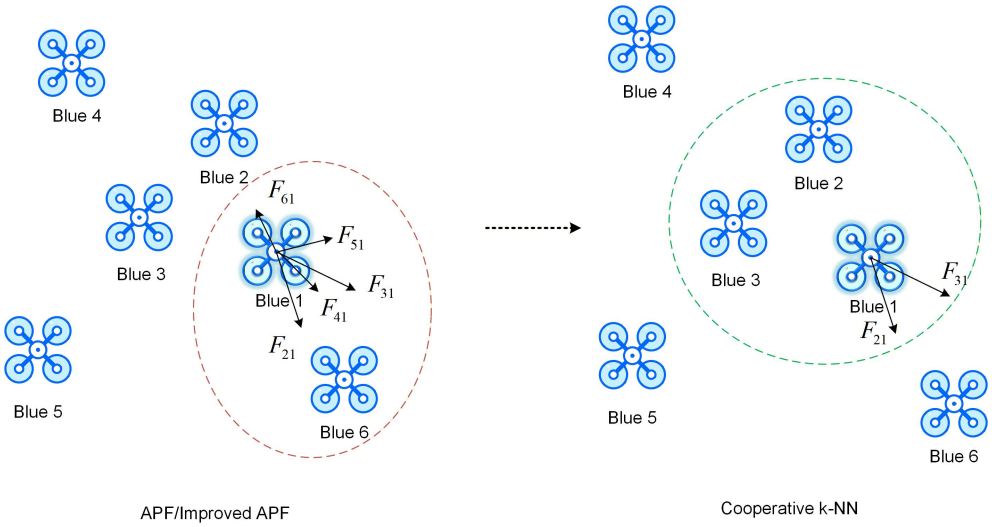


Figure 7. Comparison diagram of force analysis using APF/improved APF algorithms versus k-NN obstacle avoidance algorithm.

The cooperative obstacle avoidance algorithm based on K-nearest neighbor (K-NN) nonlinear repulsion addresses inter-vehicle collision risks during formation flight in dense airspaces. Table 2 and Figure 8 present the performance comparison among the traditional artificial potential field (APF) method, the APF with an escape mechanism (improved APF), and the proposed K-nearest neighbor (K-NN) nonlinear repulsion model (cooperative k-NN).

With task assignment and guidance modules fixed to baseline configurations, this section isolates collision avoidance as the sole variable. All experimental settings—including the four engagement scales and 50 independent repetitions per configuration—are kept identical to those described in Section 2.

Regarding the core safety metric of collision rates, the Cooperative k-NN algorithm demonstrates a significant advantage in scale adaptability, as shown in Figure 8a. It achieves optimal collision control performance across three medium-to-large-scale scenarios (30v40, 40v50, and 50v60), reducing collision rates to 0.5000, 0.3000, and 0.2833, respectively. Compared to the traditional APF's 1.0500 in the 50v60 scenario, this represents a 73.0% reduction, and a 94.0% reduction compared to the Improved APF's 4.7333 in the same scenario. While the Improved APF performs acceptably in small-to-medium scales, its performance deteriorates sharply in the 50v60 large-scale scenario, with the collision rate soaring to 4.7333. This reflects a scalability bottleneck in dense environments, where stochastic escape maneuvers appear to trigger a "domino effect." The traditional APF shows anomalous performance in the 30v40 scenario (collision rate of 1.5750) and overall instability, suggesting a lack of robustness in complex environments.

Table 2. Obstacle Avoidance Algorithm Performance Comparison.

Scale	Algorithm	Collision Rate [3]	Max Collision Risk [4]	Time (ms)	Formation Keeping [5]
15v20	APF	0.2500	0.545	0.416	0.046
	Improved APF	0.2500	0.622	0.374	0.105
	Cooperative k−NN	**0.2500**	**0.588**	**0.337**	**0.112**
30v40	APF	1.5750	0.626	0.715	0.023
	Improved APF	0.6250	0.641	0.725	0.025
	Cooperative k−NN	**0.5000**	**0.632**	**0.774**	**0.027**
40v50	APF	0.6200	0.668	0.939	0.024
	Improved APF	1.7600	0.644	0.988	0.021
	Cooperative k−NN	**0.3000**	**0.637**	**1.091**	**0.020**
50v60	APF	1.0500	0.660	1.240	0.025
	Improved APF	4.7333	0.646	1.430	0.019
	Cooperative k−NN	**0.2833**	**0.635**	**1.554**	**0.019**

[3] Refers to the average number of collision events per trial, where a collision is counted when the distance between any two UAVs falls below 0.1 m (with a 5-step deduplication window to avoid double-counting continuous contacts). [4] Collision Risk is defined per pair per time step as $Risk = \min\left(1, \max(0, 1 - (d_{ij} - d_{safe})/(d_{alert} - d_{safe}))\right)$, where $d_{safe} = 3.0$ m is the safe distance threshold and $d_{alert} = 8.0$ m is the alert distance threshold. The reported value is the mean across all UAV pairs and all time steps. [5] Formation Keeping: $1/(\text{mean}(\sigma_x, \sigma_y, \sigma_z) + 0.1)$, where σ is the spatial standard deviation at the final time step. The added constant 0.1 prevents division by zero and ensures the metric remains finite even in cases of perfect formation. Bold: highlight the proposed method.

In terms of the maximum collision risk metric, the three algorithms perform relatively similarly, maintaining a range between 0.545 and 0.678 with a variance of no more than 12.4%, indicating balanced capabilities in extreme risk control, as shown in Figure 8b. Notably, maximum collision risk shows a weak correlation with the actual collision rate; for instance, the Improved APF's maximum risk in the 50v60 scenario is only 0.646 (near the lowest value among the three algorithms at this scale, where the Cooperative k-NN achieves 0.635), yet its actual collision rate is the worst at 4.7333. This suggests that while average risks are manageable, a small number of high-risk UAVs fail to avoid collisions effectively, reflecting a deficiency in handling individual variances. In contrast, Cooperative k-NN maintains both low maximum risk and a low actual collision rate, demonstrating superior global coordination.

Regarding computational latency, the execution time for the Cooperative k-NN algorithm grows from 0.337 ms (15v20) to 1.554 ms (50v60), reflecting the increased computational demand of the k-NN coordination mechanism as the number of UAVs rises, as shown in Figure 8c. However, even at the 50v60 scale, the 1.554 ms latency fully satisfies real-time requirements, proving the algorithm's suitability for practical deployment.

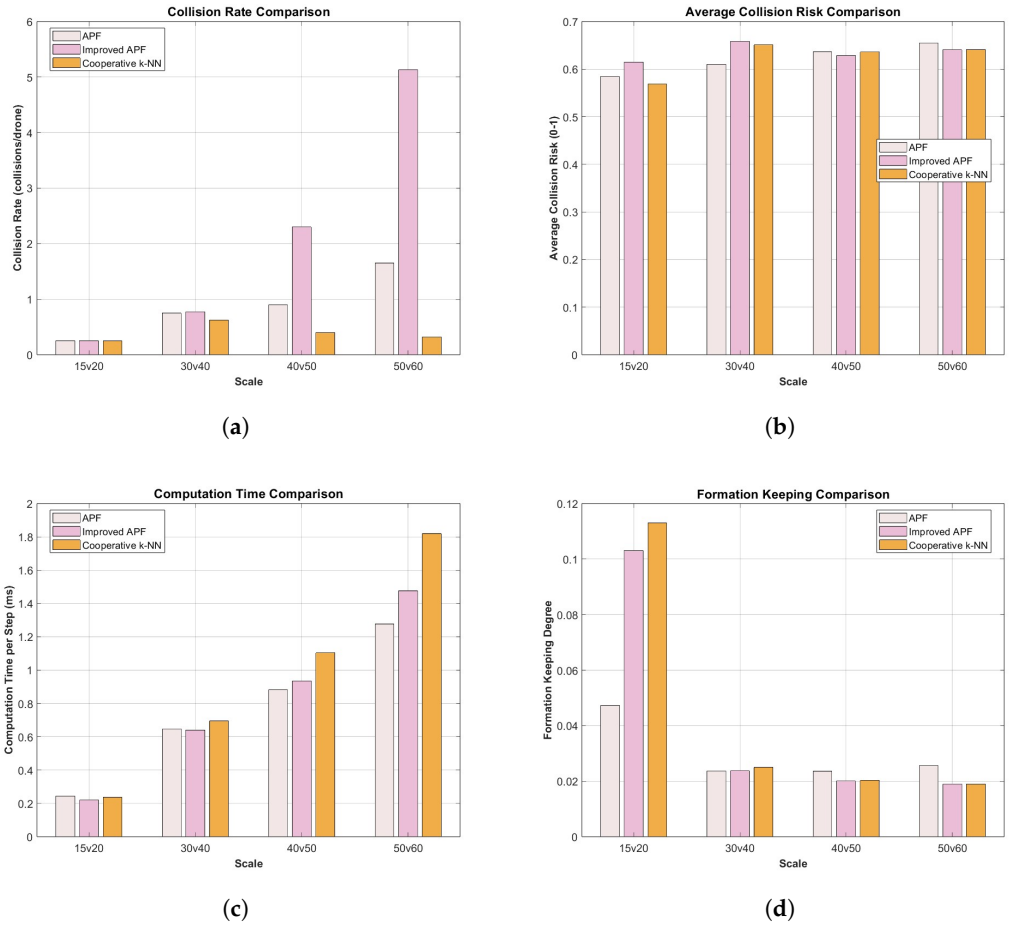


Figure 8. (**a**) Collision rate comparison. (**b**) Average collision risk comparison. (**c**) Computation time comparison. (**d**) Formation keeping comparison.

For formation maintenance, the traditional APF holds a significant advantage in the 15v20 scenario (0.046), highlighting the role of a strong centralized potential field in maintaining formation cohesion, as shown in Figure 8d. However, as the scale increases to 30v40 and beyond, the formation maintenance metrics for all three algorithms converge rapidly to a narrow range of 0.019–0.027, indicating that all algorithms face similar dispersion challenges in dense settings. This overall downward trend aligns with expected scale effects: increased UAV density leads to frequent avoidance maneuvers, forcing individuals to deviate from ideal positions to bypass obstacles and peers.

In summary, the results clearly illustrate that the Cooperative k-NN algorithm, through its k-NN coordinated avoidance mechanism, significantly improves safety performance while maintaining real-time computational efficiency. It achieves an effective unification of safety and efficiency in complex adversarial scenarios. The performance collapse of the Improved APF in large-scale scenarios suggests a need for scale-related stress testing prior to deployment, while the traditional APF remains better suited as a baseline for small-scale applications.

4. Terminal Guidance Algorithm

4.1. Pure Pursuit

In cooperative UAV interception missions, how to adjust interception strategies in real-time based on the target's motion state is the core issue for achieving efficient interception. The challenge of this problem lies in the fact that the target may perform evasive maneuvers, while the interception algorithm needs to rapidly approach the target while simultaneously avoiding collisions between the blue UAVs.

The Pure Pursuit (PP) algorithm is a classic method in the fields of missile guidance and UAV interception. The core idea of this algorithm is to continuously move toward the target's current position, and it is simple to implement.

The central principle of the Pure Pursuit algorithm is to continuously move toward the target's instantaneous position while introducing a fixed look-ahead distance to smooth the trajectory.

For the i-th blue UAV, define the unit direction vector pointing toward the target $\mathbf{p}_t$ as:

$$\mathbf{d}_t = \frac{\mathbf{p}_t - \mathbf{p}_i^b}{\|\mathbf{p}_t - \mathbf{p}_i^b\|} \tag{24}$$

The look-ahead distance is jointly determined by a fixed gain coefficient k_{gain} and a maximum look-ahead distance $L_{\max}$:

$$L = \min\left(v_i \cdot k_{gain}, L_{\max}\right), (k_{gain} = 0.8, L_{\max} = 50.0\ \text{m}) \tag{25}$$

Based on this, the look-ahead point position is calculated as:

$$\mathbf{p}_l = \mathbf{p}_i^b + \mathbf{d}_t \cdot L \tag{26}$$

The final movement direction d_p is obtained via the linear fusion of the unit direction vector pointing toward the look-ahead point

$$\mathbf{d}_l = \frac{\mathbf{p}_l - \mathbf{p}_i^b}{\|\mathbf{p}_l - \mathbf{p}_i^b\|} \tag{27}$$

and the k-nearest neighbor cooperative obstacle avoidance direction, $\mathbf{d}'_{avoid}$ (as defined in Equation (21)):

$$\mathbf{d}_p = (1 - w_{avoid}) \cdot \mathbf{d}_l + w_{avoid} \cdot \mathbf{d}'_{avoid}, (w_{avoid} = 0.3) \tag{28}$$

where $w_{avoid} \in [0, 1]$ is the obstacle avoidance weight coefficient, and the final desired heading direction $\mathbf{d}'_p$ is obtained by normalizing the synthesized vector $\mathbf{d}_p$. The position update utilizes a fixed-step kinematic model:

$$\mathbf{p}_{i,next}^b = \mathbf{p}_i^b + \mathbf{d}'_p \cdot \Delta s \tag{29}$$

where $\Delta s = \min(v_i \cdot \Delta t, L)$ is the single-step movement distance, and Δt is the simulation time step.

Despite the simplicity and theoretical maturity of the Pure Pursuit algorithm, it exhibits the following deficiencies in practical interception scenarios:

1. Static Gain Limitations: Since k_{gain} is a constant and does not adapt dynamically to changes in distance or velocity, there is a significant performance gap between long-range and short-range scenarios.

2. Lack of Predictive Capability: The algorithm lacks the ability to anticipate future states; it merely "chases" the target's tail, leading to low interception efficiency against maneuvering targets.
3. Proximity Overflight: When the separation between the blue UAV and the target is small, the look-ahead distance can become relatively too large, causing the UAV to overshoot or fly over the target.
4. Long-Range Inefficiency: When the separation between the blue UAV and the target is large, an insufficiently small look-ahead distance can result in a convoluted, "winding" path, making the actual travel distance significantly longer than the direct line-of-sight distance.

4.2. Improved Pure Pursuit Algorithm

4.2.1. Adaptive Look-Ahead Algorithm

The Adaptive Look-ahead algorithm addresses the Pure Pursuit algorithm's inability to distinguish between "near and far" scenarios by dynamically adjusting the look-ahead gain coefficient based on the distance to the target [32].

The core logic is as follows: reduce look-ahead at close range to improve precision, and increase look-ahead at long range to enhance efficiency. While this scheme effectively reduces overshooting and maintains a standardized look-ahead, its piecewise strategy remains coarse and lacks predictive capability. Fundamentally, it remains a "pursuit" rather than an "interception" method, leading to severe response lag when facing targets that perform sudden maneuvers.

4.2.2. Target Prediction Algorithm

The Target Prediction algorithm addresses the "passive response" deficiency of adaptive algorithms by introducing a target motion prediction mechanism. The core idea is to stop chasing the target's current position and instead predict its future location to perform an active interception [33,34].

The algorithm employs a linear extrapolation method to estimate the velocity vector of the red target. Let the current simulation time be t and the time step be Δt. Given the target's positions at consecutive moments, $\mathbf{p}_t(t)$ and $\mathbf{p}_t(t-\Delta t)$, the estimated target velocity is:

$$\hat{\mathbf{v}}_t = \frac{\mathbf{p}_t(t) - \mathbf{p}_t(t-\Delta t)}{\Delta t} \tag{30}$$

Based on this, the target's position at a future time $T_{predict}$ is predicted:

$$\mathbf{p}_{pred} = \mathbf{p}_t(t) + \hat{\mathbf{v}}_t \cdot T_{predict}(d) \tag{31}$$

where $d = \|\mathbf{p}_t(t) - \mathbf{p}_i^b\|$ is the current distance between the blue UAV and the target, and:

$$T_{predict}(d) = d / \max(v_i^b - 0.2v_t^r, 15) \tag{32}$$

$T_{predict}(d)$ is the prediction look-ahead time, estimated as the time required to close the relative distance d at the estimated relative speed $(v_i^b - 0.2v_t^r)$, clamped to a minimum of 15 m/s to avoid excessive prediction horizons. The longer the distance, the longer the prediction time, allowing for the early planning of the interception path. The interception direction shifts from "toward the target's current position" to "toward the predicted intercept point." It is presented as follows:

$$\mathbf{d}_{pred} = \frac{\mathbf{p}_{pred} - \mathbf{p}_i^b}{\|\mathbf{p}_{pred} - \mathbf{p}_i^b\|} \tag{33}$$

This algorithm upgrades the interception strategy from passive "pursuit" to active "interception," significantly improving path efficiency. For targets in constant-velocity or constant-acceleration linear motion, linear extrapolation provides sufficient prediction accuracy and yields excellent interception results. However, the limitation of this method is its sole reliance on first-order linear extrapolation without integrating higher-order motion models or filtered estimation. When the target suddenly turns or performs large-scale maneuvers, the predicted point may deviate severely from the actual trajectory, leading to interception failure.

4.2.3. Cooperative Optimization Algorithm

The Cooperative Optimization algorithm addresses the limitations of the aforementioned methods by proposing a two-layer optimization strategy:

1. Integration of Linear Extrapolation and Motion Trend Analysis: This combines first-order velocity estimation with an analysis of the target's movement patterns, allowing the system to better anticipate maneuvers rather than relying solely on current velocity.
2. Dynamic Parameter Adaptation Based on UAV Quantity and Distance: All algorithmic parameters—including gain coefficients and look-ahead windows—are dynamically adjusted in real-time according to the swarm size and the relative distance to the target, ensuring optimal performance across varying mission scales and proximity levels.

Specifically, by replacing $\mathbf{d}_l$ with $\mathbf{d}_{pred}$, we obtain:

$$\mathbf{d}_{coop} = (1 - w_{coop}) \cdot \mathbf{d}_{pred} + w_{coop} \cdot \mathbf{F}'_{avoid} \tag{34}$$

The final desired heading direction $\mathbf{d}'_{coop}$ is obtained by normalizing the synthesized vector $\mathbf{d}_{coop}$. Simultaneously, the weight coefficients are dynamically determined based on the number of UAVs n and the distance to the target d:

$$w_{coop}(n, d) = \begin{cases} (0.18, 0.10), & \text{if } n \leq 20 \\ (0.20, 0.12), & \text{if } 20 < n \leq 35 \\ (0.22, 0.14), & \text{if } n > 35 \end{cases} \tag{35}$$

where the first value in the parentheses represents the short-range weight ($d < 25$ m) and the second value represents the long-range weight ($d \geq 25$ m).

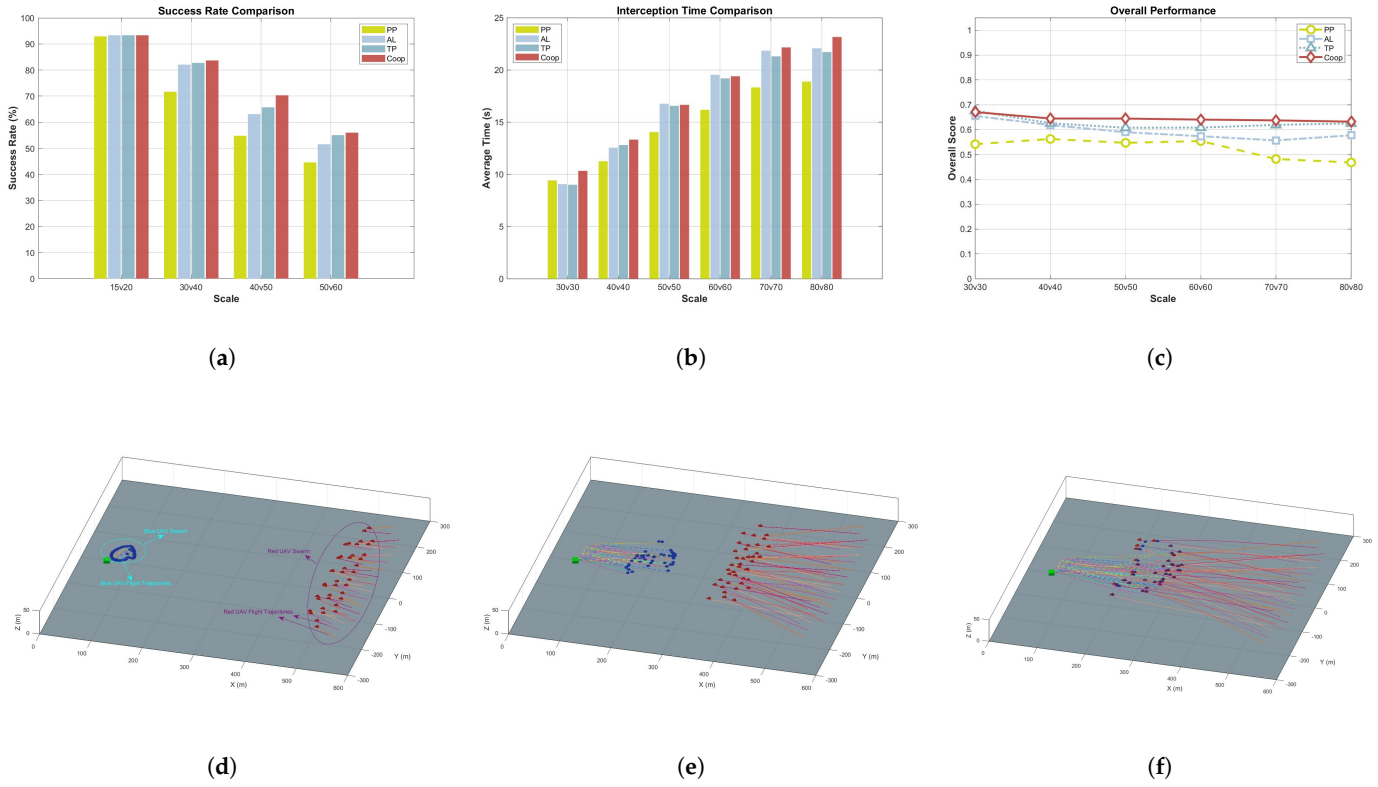
The adaptive pure pursuit guidance algorithm integrating target motion prediction addresses tracking lag and target loss issues during interception of highly maneuverable targets. Table 3 and Figure 9a–c present the performance comparison among the pure pursuit (PP), adaptive look-ahead (AL), target prediction (TP), and cooperative (Coop) algorithms.

In this section, the task allocation module is fixed as the load-balancing greedy algorithm described in Section 3, and the inter-UAV obstacle avoidance module is fixed as the cooperative k-nearest neighbor mechanism described in Section 4, with the guidance algorithm serving as the sole variable. All experimental settings—including the four engagement scales (15v20, 30v40, 40v50, and 50v60) and 50 independent repetitions for each configuration—remain strictly consistent with those previously mentioned. Consequently, the performance metrics reported below reflect the marginal contribution of the guidance layer under a unified underlying architecture.

Table 3. Interception Algorithm Performance Comparison.

Scale	Algorithm	Success Rate (%)	Avg Time (s)	Overall Score [6]
15v20	PP	92.93	6.53	0.697
	AL	93.33	6.38	0.707
	TP	93.33	6.34	0.709
	Coop	**93.33**	**6.24**	**0.713**
30v40	PP	71.67	9.42	0.600
	AL	82.07	9.15	0.694
	TP	**82.73**	**9.08**	**0.705**
	Coop	83.67	9.65	0.693
40v50	PP	54.80	11.49	0.546
	AL	63.10	12.42	0.604
	TP	65.70	12.08	0.638
	Coop	**70.30**	**12.63**	**0.670**
50v60	PP	44.60	14.29	0.558
	AL	51.56	16.71	0.589
	TP	55.04	16.25	0.639
	Coop	**56.00**	**16.50**	**0.654**

[6] Overall Score = 0.7 × (Success Rate/Max Success Rate) + 0.3 × ((PP Time − Algorithm Time)/PP Time). Bold: highlight the proposed method.



(**a**) (**b**) (**c**)

(**d**) (**e**) (**f**)

Figure 9. (**a**) Success rate comparison. (**b**) Interception time comparison. (**c**) Overall performance. (**d**–**f**) Visualization process of red and blue UAVs in the simulation experiment: (**d**) initialization; (**e**,**f**) cooperative interception process.

In terms of the success rate metric, the performance of all algorithms exhibits a downward trend as the scenario scale increases, which aligns with expectations for multi-target, high-density confrontations where increased target numbers and airspace congestion lead to heightened decision-making complexity and collision risk for each interceptor, as shown in Figure 9a. At the small scale (15v20), the four algorithms perform similarly with success rates exceeding 92% and differences of less than 0.5%, suggesting that guidance strategy choice has limited impact in sparse airspace. However, starting at the 30v40 scale,

the proposed Cooperative Interception (Coop) algorithm takes the lead with an 83.67% success rate, while the Pure Pursuit (PP) algorithm drops to 71.67%, revealing the latter's fundamental lack of predictive and cooperative mechanisms. This advantage expands in the 40v50 scenario, where Coop reaches 70.30%, outperforming the Target Prediction (TP), Adaptive Look-ahead (AL), and PP algorithms by 4.6, 7.2, and 15.5 percentage points, respectively. Notably, in the 50v60 scenario, Coop maintains a slight lead at 56.00% with TP following closely at 55.04%; this indicates that while the cooperative mechanism provides the primary performance gain in most medium-to-large scales, the contribution of target motion prediction becomes increasingly critical in extremely dense environments where proactive positioning is needed to offset the delays caused by frequent obstacle avoidance maneuvers.

The "Avg Time" in the table represents the cumulative total duration for all blue UAVs, as shown in Figure 9b. Taking the largest scale (50v60) as an example, the Coop algorithm's total time of 16.50 s, when distributed across 50 UAVs, results in an average per-UAV time of approximately 330 ms, with single-step computational overhead remaining at the millisecond level. The total time differences between algorithms (e.g., TP saving 0.25 s compared to Coop) translate to only about 5 ms per UAV, indicating no substantive difference in real-time performance; thus, all algorithms meet the stringent timing requirements for onboard real-time interception. The sub-linear growth of total time relative to scale validates the design goal of decoupling the k-NN obstacle avoidance mechanism from swarm size. In terms of the comprehensive score (a weighted fusion of success rate and time efficiency), the Coop algorithm achieved the highest scores in the 15v20, 40v50, and 50v60 scenarios, while the TP algorithm held a slight lead in the 30v40 scale with a score of 0.705 (compared to Coop's 0.693). This phenomenon reveals the differing scale-applicability of the two improvement strategies: the target prediction mechanism provides significant interception efficiency gains at a lower computational cost in medium-scale scenarios, whereas the cooperative interception mechanism better leverages its global coordination advantages in large-scale, high-density scenarios by dynamically balancing the weights of obstacle avoidance and interception.

Overall, the proposed Cooperative Interception (Coop) algorithm achieves optimal or suboptimal comprehensive performance across the vast majority of scales, demonstrating the most robust guidance capability particularly in large-scale confrontations involving more than 40 blue UAVs, as shown in Figure 9c. As a baseline, the Pure Pursuit (PP) algorithm is only engineering-feasible in small-scale scenarios, as its performance degrades most severely with increasing scale. The Adaptive Look-ahead (AL) and Target Prediction (TP) algorithms, as single-dimension improvement schemes, perform well within specific scale intervals but lack the global coordination provided by a cooperative obstacle avoidance mechanism, which limits their performance ceiling in large-scale dense environments.

In the simulation experiments illustrated in Figure 9d–f, the red intruder trajectories are pre-defined using cubic Bézier curves and arranged in a wedge formation. To ensure the statistical validity of the evaluation and to cover a diverse set of engagement geometries, the specific Bézier control points for each red UAV are randomly regenerated at the beginning of every independent experimental repetition. Consequently, the exact flight path of each red intruder varies across the 50 runs, while the macroscopic wedge-shaped formation structure is preserved. This design introduces controlled variability into the scenario, allowing the reported performance metrics to reflect robustness against deployment uncertainty rather than overfitting to a single deterministic trajectory.

The blue interceptor swarm is initialized at ground level and takes off sequentially to ascend and assemble into a circular formation at a predetermined altitude. Once formation assembly is completed, each blue UAV receives its assigned red target from the task alloca-

tion module and initiates the cooperative interception maneuver. The entire process—from takeoff, formation assembly, target assignment, to terminal guidance—is simulated within a unified hierarchical framework. The initial deployment of both teams is scaled with the swarm size to preserve a constant nominal spatial density, thereby guaranteeing fair cross-scale comparisons.

5. Results

5.1. Simulation Configuration

5.1.1. Blue UAV Configuration

The defending (blue) UAVs are configured with the following parameters:

- **Flight speed:** Each blue UAV travels at a constant speed of 25 m/s throughout the interception phase—this is a fixed cruise speed, not an adjustable maximum, and no acceleration or deceleration is modeled. The motion follows the fixed-step kinematic model in Equation (29).
- **Initial deployment and takeoff:** All blue UAVs start at ground level ($z = 0$) and ascend sequentially (with a staggered takeoff interval of 0.1 s per drone) to assemble into a circular holding formation at an altitude of 20 m. The formation radius is scaled with the swarm size to maintain constant spatial density.
- **Interception determination:** An engagement is deemed successful when a blue UAV approaches within a physical collision threshold of 0.1 m of its assigned red target, resulting in deterministic neutralization. No self-destruct or proximity-fuze mechanism is modeled.
- **Post-engagement behavior:** After a successful engagement, the blue UAV halts at the interception point and is not reassigned to other targets. Each blue UAV engages at most one red target per trial.
- **Situational awareness:** Each blue UAV has perfect self-localization and receives the instantaneous three-dimensional position of every red UAV and every friendly blue UAV from the defense coordinator at each time step via a centralized, zero-latency broadcast channel. The relative distance and direction to the assigned target are computed locally from the position difference. Although the blue UAV is not directly provided with the red target's velocity vector, it can estimate the velocity from consecutive position observations via finite difference, $\hat{\mathbf{v}}_t = (\mathbf{p}_t(t) - \mathbf{p}_t(t - \Delta t))/\Delta t$, as detailed in Section 4.2.2. This estimated velocity is used by the predictive guidance algorithms (TP and Coop) for target motion prediction.

5.1.2. Red UAV Configuration

The attacking (red) UAVs are configured as follows:

- **Flight speed:** Each red UAV travels at a constant speed of 30 m/s along its predetermined path. No speed variation or acceleration is modeled.
- **Flight path definition:** Red trajectories are pre-programmed using cubic Bézier curves before simulation execution. Each red UAV flies from a randomly generated start position toward a protected asset located at $[50, 0, 0]$ m (scaled with swarm size). The Bézier control points are randomly regenerated at the beginning of each independent trial, introducing controlled variability while preserving the macroscopic wedge formation structure.
- **Maneuverability:** Red UAVs possess no situational awareness of the defenders and execute no evasive, reactive, or cooperative maneuvers. Their trajectories are entirely predetermined and remain unchanged regardless of blue UAV actions.

- **Post-engagement behavior:** Upon being intercepted, a red UAV stops at the interception point and remains stationary for the remainder of the trial. If a red UAV reaches the protected asset without being intercepted, it also halts.

5.1.3. Simulation Environment

The simulation environment parameters common to all three algorithmic layers are as follows:

- **Time step:** The simulation advances with a fixed time step of $\Delta t = 0.1$ s.
- **Spatial scale:** The simulation environment is scaled with the square root of the number of blue UAVs, scale $= \sqrt{n/30}$, to preserve a constant nominal spatial density across all engagement scales. At the reference scale ($n = 30$), the protected asset is located at $[50, 0, 0]$ m, and the red UAVs are initialized at approximately $x = 500$ m, traversing cubic Bézier paths toward the protected asset over a longitudinal extent of roughly 450 m. The blue UAV formation altitude is 20 m at the reference scale, and inter-drone spacing within both formations is set to 15 m. All spatial coordinates scale proportionally with $\sqrt{n}$, ensuring that the per-drone operating area remains constant across different swarm sizes.
- **Trial configuration:** Four representative engagement scales are evaluated: 15 vs. 20, 30 vs. 40, 40 vs. 50, and 50 vs. 60 (red vs. blue). In all scales, the defending (blue) UAVs moderately outnumber the attacking (red) UAVs by a ratio of approximately 1.3:1 to 1.4:1 (blue to red). This numerical advantage allows the framework to allocate surplus defensive resources while testing the load-balancing mechanism's ability to distribute interceptors efficiently across targets.
- **Simulation duration:** Each trial runs until all red UAVs have either been intercepted or have reached the protected asset, with the maximum simulation time determined by the longest Bézier path length divided by the red flight speed.

The constant-speed disadvantage of the blue defenders (25 m/s vs. red's 30 m/s) is a deliberate design choice that establishes the algorithmic motivation of this work. Under the constant-speed model, a slower pursuer cannot close the distance to a faster target through direct chase alone; successful interception necessarily requires predictive guidance that anticipates the target's future position. This constraint renders pure pursuit (PP) geometrically incapable of achieving interception against a faster intruder and establishes the rationale for the target prediction (TP) and cooperative interception (Coop) algorithms. The numerical advantage of approximately 1.3:1 to 1.4:1 (blue to red) offsets the speed deficit by enabling multi-directional bracketing—multiple defenders can converge on a single intruder from different azimuths, reducing the time-to-intercept through geometric cooperation rather than raw speed. Together, the speed disparity and numerical surplus create a tactically meaningful benchmark: defensive success depends on algorithmic coordination and predictive accuracy rather than kinematic superiority, ensuring that the reported performance metrics genuinely reflect the efficacy of the proposed hierarchical framework.

5.2. Task Allocation Performance

Table 1 demonstrates the effectiveness of the Load-Balanced Hungarian algorithm. Across all scales, LB-Hungarian achieves load standard deviations of 0.404–0.488, compared to 1.414–1.826 for the standard Hungarian algorithm. The maximum-to-minimum load ratio is consistently maintained at 2.0 for LB-Hungarian, versus 6.0–11.0 for the baseline. These results confirm that the proposed multi-dimensional threat assessment and nonlinear load penalty mechanism effectively resolve the resource idling and target overloading issues inherent in traditional one-to-one allocation. The total flight distance penalty incurred by LB-Hungarian remains below 3%, which is tactically acceptable given the substantial load-balancing gains.

5.3. Obstacle Avoidance Performance

Table 2 compares the collision rates and computational metrics of three obstacle avoidance strategies. The Cooperative k-NN algorithm achieves the lowest collision rates in three of four medium-to-large-scale scenarios (30v40, 40v50, 50v60), reducing the collision rate to 0.2833 at the 50v60 scale—a 73.0% reduction compared to traditional APF and a 94.0% reduction compared to Improved APF. Computational latency remains at millisecond level (1.554 ms at 50v60), fully satisfying real-time requirements. Notably, the Improved APF exhibits severe performance collapse at large scales (collision rate of 4.7333 at 50v60), while Cooperative k-NN maintains robust scalability.

5.4. Interception Guidance Performance

Table 3 presents the success rates and overall scores of four guidance algorithms. The proposed Coop algorithm achieves the highest success rates in three of four scenarios (15v20: 93.33%, 40v50: 70.30%, 50v60: 56.00%) and attains top overall scores at these scales. At the 30v40 scale, the TP algorithm leads with 82.73% success rate, while Coop follows closely at 83.67%. The sub-linear growth of computational time with scale validates the k-NN decoupling design. Even at the 50v60 scale, the average per-UAV computation time remains at approximately 330 ms per trial, confirming real-time feasibility.

6. Discussion

The experimental results across all three layers corroborate the effectiveness of the proposed hierarchical interception framework. The load-balanced Hungarian algorithm ensures equitable resource utilization without sacrificing overall interception efficiency. The Cooperative k-NN obstacle avoidance mechanism reduces collision risk by over 70% in dense scenarios while maintaining millisecond-level computational latency. The adaptive predictive guidance algorithm shifts the interception paradigm from passive pursuit to active interception, yielding substantial success rate improvements in large-scale confrontations.

Although UAVs are inherently dynamic systems subject to inertial, aerodynamic, and actuation constraints, the proposed framework adopts a point-mass kinematic model (Assumption 1 in Section 1) to serve as an idealized upper-bound benchmark. This modeling choice isolates the algorithmic properties of the hierarchical interception architecture—task allocation, cooperative obstacle avoidance, and predictive guidance—from low-level dynamic coupling, providing an upper-bound performance benchmark that is independent of specific platform characteristics. Recent advances in dynamic system control, such as the finite-time neural adaptive control framework [35] and the fast finite-time dynamic surface control method [36] developed for electro-hydraulic servo systems, demonstrate effective handling of parametric uncertainty, input delay, and nonlinearity through Lyapunov-based backstepping designs. These techniques are, in principle, transferable to the low-level attitude and velocity control loops of individual UAVs within the proposed hierarchy. Integrating such methods to bridge the gap between kinematic planning and dynamic execution represents a promising direction for extending the framework toward real-world deployment, where physical effects such as inertia, wind disturbance, and actuator saturation must be explicitly accommodated.

It should be acknowledged that the adaptive parameters and fusion weight coefficients in this study are based on extensive preliminary experiments and physical intuition. Systematic parameter optimization methods for dynamical systems have been developed in the literature—for instance, Nie et al. [19] proposed a genetic algorithm-based framework for stability analysis of time-delay systems, demonstrating automated tuning of system parameters to meet prescribed stability constraints. While such evolutionary computation techniques (e.g., genetic algorithms (GA), particle swarm optimization (PSO), and Bayesian

optimization) could, in principle, be applied to optimize the empirical parameters in the present framework (e.g., the scale-adaptive coefficients α_{scale}, w_{dist}, w_{angle}, and λ_{scale} in Section 2, or the obstacle avoidance parameters k, d_r, and λ in Section 3), the primary objective of this paper is to establish and validate the hierarchical architecture itself rather than to pursue parameter-level optimality. The consistent performance improvements across multiple scales (from 15v20 to 50v60) indirectly validate the robustness of the proposed parameterization scheme, and incorporating systematic optimization techniques remains a worthwhile direction for future refinement.

Several recent studies have addressed multi-UAV interception from complementary perspectives. Wang et al. [11] proposed an optimal guidance framework that integrates pure pursuit guidance, Kuhn–Munkres optimal matching, and virtual force potential fields for collision avoidance—effectively combining task allocation, guidance, and collision avoidance within a unified architecture. Xia et al. [15] modeled UAV swarms as mobile adaptive networks and developed a distributed modularized framework for collaborative multi-target tracking, decision-making, and motion control in dynamic confrontation scenarios. Taranov [16] introduced a multilayer UAV swarm formation specifically designed for target interception missions. Hu and Wei [17] formulated quadrotor interception as a zero-sum differential game and derived the guidance law accordingly, with sliding mode control employed for attitude tracking. Li et al. [18] applied multi-agent reinforcement learning (actor-critic with centralized training and decentralized execution) to collaborative decision-making in multi-UAV air combat. Wen et al. [20] proposed a distributed target-fencing control strategy for UAV swarms under saturation attack missions, using consensus theory with an auxiliary state to compensate for the absence of speed measurements. Compared with these approaches, the proposed framework is distinguished by its introduction of a dedicated nonlinear load-balancing mechanism within the task allocation layer and a k-NN cooperative obstacle avoidance module that reduces computational complexity from $O(n^2)$ to $O(kn)$, coupled with an adaptive predictive guidance law that explicitly accounts for cross-layer coupling: the obstacle avoidance weight w_{avoid} is dynamically adjusted based on the formation scale inherited from the task allocation layer, while the predictive guidance operates within the safe separation guaranteed by the collision avoidance module. This cross-layer coordination enables robust and scalable performance across engagement scales ranging from 15v20 to 50v60. It should be noted, however, that direct quantitative comparison with the aforementioned methods is complicated by differences in simulation assumptions, scenario scales, and evaluation metrics; establishing a standardized benchmarking platform for multi-UAV interception remains an important community objective.

Future research will focus on the following directions. First, a comprehensive sensitivity analysis and systematic parameter optimization will be conducted—employing GA, PSO, or Bayesian optimization to automatically calibrate the scale-adaptive coefficients and fusion weights, thereby reducing reliance on empirical tuning. Second, the validation pipeline will be advanced from purely numerical simulation to software-in-the-loop (SITL) and hardware-in-the-loop (HITL) testing, incorporating realistic aerodynamic models, sensor noise, and communication delays to bridge the gap between the kinematic assumptions adopted herein and physical dynamics. Third, outdoor flight experiments with a prototype multi-UAV platform will be carried out to validate the framework's robustness and practicality under real-world conditions, addressing the full spectrum of challenges—including localization uncertainty, wind disturbances, and computational constraints—that are abstracted away in the current simulation. Fourth, reinforcement learning-based online adaptation will be explored to enable the guidance and obstacle avoidance policies to respond to unforeseen intruder behaviors that deviate from the pre-programmed trajectory

model assumed in this study. Together, these steps will accelerate the translation of the proposed hierarchical interception framework from simulation-based proof-of-concept toward operational deployment.

7. Conclusions

This paper proposed a hierarchical interception framework that integrates load-balanced greedy assignment, cooperative k-NN collision avoidance, and adaptive predictive pure-pursuit guidance. Extensive simulations across four swarm scales confirm that the framework achieves balanced task distribution, substantially reduced collision rates, and improved interception success rates while maintaining real-time feasibility, offering a robust and scalable solution for defending against large-scale UAV intrusions in complex urban environments.

Author Contributions: Funding acquisition, Conceptualization, L.Z.; Software, Validation and Writing—original draft, Y.W.; Investigation, J.L.; Formal analysis, Project administration, and Writing—review and editing, Y.L. and R.G.; Methodology, L.Z. and R.G. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The data and code supporting the conclusions of this article can be obtained from the corresponding author upon reasonable request.

DURC Statement: Current research is limited to the field of multi-UAV cooperative task allocation, collision avoidance, and guidance algorithms, which is beneficial for providing security solutions in civilian domains, such as the protection of critical infrastructure (e.g., nuclear facilities and major urban ports) against unauthorized drone intrusion and interference, and does not pose a threat to public health or national security. Authors acknowledge the dual-use potential of the research involving UAV swarm interception technologies and confirm that all necessary precautions have been taken to prevent potential misuse. As an ethical responsibility, authors strictly adhere to relevant national and international laws about DURC. Authors advocate for responsible deployment, ethical considerations, regulatory compliance, and transparent reporting to mitigate misuse risks and foster beneficial outcomes.

Acknowledgments: The authors extend their sincere gratitude to the anonymous reviewers and the editor for their insightful comments and valuable suggestions, which have significantly contributed to enhancing the quality of this research.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Stanev, H.; Mitev, S. Organisational and Technical Challenges for Conflict-free use of Civil Airport Airspace by Drones. In *Proceedings of the 2025 33rd Telecommunications Forum (TELFOR), Belgrade, Serbia*; IEEE: Piscataway, NJ, USA, 2025; pp. 1–4.
2. Zhang, T.; Lu, R.; Yang, X.; Xie, X.; Fan, J.; Tang, B. UAV Hunter: A Net-Capturing UAV System with Improved Detection and Tracking Methods for Anti-UAV Defense. *Drones* **2024**, *8*, 573. [CrossRef]
3. Geng, X.; Tang, K.; Guo, Y.; Zhang, L.; Wu, W.; Ma, T. A Distributed UAV Swarm Countermeasure Method Based on GNSS Spoofing. *IEEE Internet Things J.* **2025**, *12*, 33455–33467. [CrossRef]
4. Liao, L.; Huang, X.; Xie, F. Development Status and Operation Analysis of Laser Weapon in Anti-Drone Warfare. In *Proceedings of the 2023 IEEE International Conference on Unmanned Systems (ICUS), Hefei, China*; IEEE: Piscataway, NJ, USA, 2023; pp. 305–310.
5. Chiper, F.-L.; Martian, A.; Vladeanu, C.; Marghescu, I.; Craciunescu, R.; Fratu, O. Drone Detection and Defense Systems: Survey and a Software-Defined Radio-Based Solution. *Sensors* **2022**, *22*, 1453. [CrossRef]
6. Wang, J.; Liu, Y.; Song, H. Counter-Unmanned Aircraft System(s) (C-UAS): State of the Art, Challenges, and Future Trends. *IEEE Aerosp. Electron. Syst. Mag.* **2021**, *36*, 4–29. [CrossRef]
7. Shen, Q.; Gong, S.; Varlamov, O.O. Design of the "Sky Watcher" Intelligent Anti-Drone System. In *Proceedings of the 2024 2nd Asian Aerospace and Astronautics Conference (AAAC), Nanjing, China*; IEEE: Piscataway, NJ, USA, 2024; pp. 39–44.

8. Wang, L.; Luo, J.; Li, Z.; Wang, M.; Li, R.; Xu, W.; He, F.; Xu, B. Thinking about anti-drone strategies. In *Proceedings of the 2024 3rd International Conference on Artificial Intelligence, Internet of Things and Cloud Computing Technology (AIoTC), Wuhan, China*; IEEE: Piscataway, NJ, USA, 2024; pp. 53–56.
9. Zhao, X.; Zhou, H. Distributed Collaborative Control Technology in Multiple Unmanned Aerial Vehicle Operations. In *Proceedings of the 2023 5th International Conference on Robotics, Intelligent Control and Artificial Intelligence (RICAI), Hangzhou, China*; IEEE: Piscataway, NJ, USA, 2023; pp. 393–396.
10. Yang, F.; Lu, Q.; Chen, M.; Lin, J.; Zhang, B. Distributed Autonomous Formation Tracking for UAV Swarm in Complex Environments. In *Proceedings of the 2025 44th Chinese Control Conference (CCC), Chongqing, China*; IEEE: Piscataway, NJ, USA, 2025; pp. 4692–4697.
11. Wang, X.; Tan, G.; Dai, Y.; Lu, F.; Zhao, J. An Optimal Guidance Strategy for Moving-Target Interception by a Multirotor Unmanned Aerial Vehicle Swarm. *IEEE Access* **2020**, *8*, 121650–121664. [CrossRef]
12. Balampanis, F.; Aguiar, A.P.; Maza, I.; Ollero, A. Path tracking for waypoint lists based on a pure pursuit method for fixed wing UAS. In *Proceedings of the 2017 Workshop on Research, Education and Development of Unmanned Aerial Systems (RED-UAS), Linköping, Sweden*; IEEE: Piscataway, NJ, USA, 2017; pp. 55–59.
13. Yamasaki, T.; Balakrishnan, S.N. Sliding mode based pure pursuit guidance for UAV rendezvous and chase with a cooperative aircraft. In *Proceedings of the 2010 American Control Conference, Baltimore, MD, USA*; IEEE: Piscataway, NJ, USA, 2010; pp. 5544–5549.
14. Kim, M.; Kim, Y. Multiple UAVs Nonlinear Guidance Laws for Stationary Target Observation with Waypoint Incidence Angle Constraint. *Int. J. Aeronaut. Space Sci.* **2013**, *14*, 67–74. [CrossRef]
15. Xia, W.; Zhou, Z.; Jiang, W.; Zhang, Y. Dynamic UAV Swarm Confrontation: An Imitation Based on Mobile Adaptive Networks. *IEEE Trans. Aerosp. Electron. Syst.* **2023**, *59*, 7183–7202. [CrossRef]
16. Taranov, D.V. Target Interception Using a Multilayer UAV Swarm Formation. In *Proceedings of the 2025 IEEE 8th International Conference on Methods and Systems of Navigation and Motion Control (MSNMC), Kyiv, Ukraine*; IEEE: Piscataway, NJ, USA, 2025; pp. 1–5.
17. Hu, C.; Wei, Y. Interception guidance and control algorithm for quadrotor unmanned aerial vehicle (UAV) against maneuvering targets. In *Proceedings of the CSAA/IET International Conference on Aircraft Utility Systems (AUS 2024), Xi'an, China*; IEEE: Piscataway, NJ, USA, 2024; pp. 1656–1662.
18. Li, S.; Jia, Y.; Yang, F.; Qin, Q.; Gao, H.; Zhou, Y. Collaborative Decision-Making Method for Multi-UAV Based on Multiagent Reinforcement Learning. *IEEE Access* **2022**, *10*, 91385–91396. [CrossRef]
19. Nie, Y.; Zhao, C.; Shi, K.; Chen, Y. Novel Stability Analysis for Time-Delay Systems via Two New Lemmas and Genetic Algorithm. *Chaos Solitons Fractals* **2026**, *208*, 118232. [CrossRef]
20. Wen, L.; Zhen, Z.; Tao, C.; Ding, J. Distributed Cooperative Strategy of UAV Swarm Without Speed Measurement Under Saturation Attack Mission. *IEEE Trans. Aerosp. Electron. Syst.* **2024**, *60*, 4518–4529. [CrossRef]
21. Zhou, L.; Leng, S.; Liu, Q.; Wang, Q. Intelligent UAV Swarm Cooperation for Multiple Targets Tracking. *IEEE Internet Things J.* **2022**, *9*, 743–754. [CrossRef]
22. Wei, K.; Huang, K.; Wu, Y.; Li, Z.; He, H.; Zhang, J.; Chen, J.; Guo, S. High-Performance UAV Crowdsensing: A Deep Reinforcement Learning Approach. *IEEE Internet Things J.* **2022**, *9*, 18487–18499. [CrossRef]
23. Yue, L.; Yang, R.; Zuo, J.; Zhang, Y.; Li, Q.; Zhang, Y. Unmanned Aerial Vehicle Swarm Cooperative Decision-Making for SEAD Mission: A Hierarchical Multiagent Reinforcement Learning Approach. *IEEE Access* **2022**, *10*, 92177–92191. [CrossRef]
24. Kuhn, H.W. The Hungarian method for the assignment problem. *Nav. Res. Logist. Q.* **1955**, *2*, 83–97. [CrossRef]
25. Jiao, Z.; Yao, P.; Zhang, J.; Zhong, Y.; Wang, X. MAV/UAV task coalition phased-formation method. *J. Syst. Eng. Electron.* **2019**, *30*, 402–414.
26. Saad, W.; Han, Z.; Basar, T.; Debbah, M.; Hjorungnes, A. A selfish approach to coalition formation among unmanned air vehicles in wireless networks. In *Proceedings of the 2009 International Conference on Game Theory for Networks, Istanbul, Turkey*; IEEE: Piscataway, NJ, USA, 2009; pp. 259–267.
27. Hong, Y.; Jung, S.; Kim, S.; Cha, J. Extensions of Receding Horizon Task Assignment for Area Coverage in Dynamic Environments. *IEEE Trans. Aerosp. Electron. Syst.* **2023**, *59*, 3114–3128. [CrossRef]
28. Zhang, R.; Chen, X.; Li, M. Multi-UAV cooperative task assignment based on multi-strategy improved DBO. *Clust. Comput.* **2025**, *28*, 195. [CrossRef]
29. Khatib, O. Real-Time Obstacle Avoidance for Manipulators and Mobile Robots. In *Proceedings of the 1985 IEEE International Conference on Robotics and Automation, St. Louis, MO, USA*; IEEE: Piscataway, NJ, USA, 1985; Volume 2, pp. 500–505.
30. Katsev, M.; Yershova, A.; Tovar, B.; Ghrist, R.; LaValle, S.M. Mapping and Pursuit-Evasion Strategies For a Simple Wall-Following Robot. *IEEE Trans. Robot.* **2011**, *27*, 113–128. [CrossRef]
31. Zafar, M.; Khan, R.A.; Batool, F.; Yaqoot, Y.; Guo, Z.; Litvinov, M.; Fedoseev, A.; Tsetserukou, D. SwarmVLM: VLM-Guided Impedance Control for Autonomous Navigation of Heterogeneous Robots in Dynamic Warehousing. In *Proceedings of the 2025 IEEE International Conference on Robotics and Biomimetics (ROBIO), Chengdu, China*; IEEE: Piscataway, NJ, USA, 2025; pp. 770–775.

32. Ahn, J.; Shin, S.; Kim, M.; Park, J. Accurate Path Tracking by Adjusting Look-Ahead Point in Pure Pursuit Method. *Int. J. Automot. Technol.* **2021**, *22*, 119–129. [CrossRef]
33. Wang, Y.; Li, D.; Zhuang, X.; Wang, Y.; Yang, S.; Wu, R. An adaptive path tracking controller for autonomous vehicles based on the Pure Pursuit algorithm. In *Proceedings of the 2023 China Automation Congress (CAC), Chongqing, China*; IEEE: Piscataway, NJ, USA, 2023; pp. 7508–7512.
34. Yick, J.; Mukherjee, B.; Ghosal, D. Analysis of a prediction-based mobility adaptive tracking algorithm. In *Proceedings of the 2nd International Conference on Broadband Networks, Boston, MA, USA*; IEEE: Piscataway, NJ, USA, 2005; Volume 1, pp. 753–760.
35. Li, S.; Yan, K.; Xie, Y.; Zhong, Q.; Yang, J.; Liao, D. Finite-Time Neural Adaptive Control of Electro-Hydraulic Servo Systems with Minimal Input Delay and Parametric Uncertainty via Padé Approximation. *Mathematics* **2026**, *14*, 1368. [CrossRef]
36. Li, S.; Yan, Y.; Yu, Y.; Zhong, Q.; Hua, L.; Liao, D. Fast Finite-Time Position Tracking Control of Electro-Hydraulic Servo Systems with Parametric Uncertainty via Dynamic Surface and Neural Adaptive Method. *Mathematics* **2026**, *14*, 1551. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.